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Internship Project

Design, Analysis, and Modeling of a CNC Milling Spindle System

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Abstract

This project presents the design, analysis, and modeling of a CNC milling spindle system as an integrated mechatronic system. The spindle is designed for high-speed operation with a maximum target speed of 5000 RPM, requiring consideration of mechanical strength, bearing support, motor selection, variable-speed operation, sensing, and control. The mechanical design includes the spindle shaft, bearing arrangement, housing, mounting components, and tool interface. Computer-aided design is used to develop the spindle assembly, while analytical hand calculations and a beam finite-element model are used to evaluate the structural performance of the spindle shaft, complemented by analytical checks of the housing and drawbar hardware.

The electrical and power electronic aspects of the system are incorporated through the selection and modeling of a suitable AC spindle motor and variable frequency drive (VFD). The drive system is represented using an AC-DC-AC architecture consisting of a rectifier, DC link, and three-phase inverter, with pulse width modulation (PWM) used for motor control. A mathematical model of the motor-spindle system is developed based on the rotational dynamics of the system and is used to study the spindle response to speed commands and load disturbances. Closed-loop PID control is implemented using spindle speed feedback to regulate the operating speed.

The automatic tool changer (ATC) and pneumatic drawbar are included as functional components of the overall spindle system. Their detailed mechanical and pneumatic analysis is outside the scope of the project; instead, they are incorporated into the supervisory control sequence. A programmable logic controller (PLC) is modeled as a supervisory layer responsible for sequencing and interlocking the spindle and tool-release operation. In particular, the PLC prevents tool release until the spindle has reached a safe standstill condition. Simulation results are used to evaluate the mechanical performance, motor-spindle dynamic response, PID speed regulation, and PLC-based supervisory interlocking. The project therefore integrates mechanical design, electrical drives, sensors, feedback control, and industrial automation into a unified CNC spindle mechatronic system.

Keywords: CNC Spindle, Automatic Tool Changer, Pneumatic Drawbar, Finite Element Analysis (FEA), Variable Frequency Drive (VFD), Pulse Width Modulation (PWM), Sinusoidal Pulse Width Modulation (SPWM), PID Control, PLC Interlocking, Mechatronics

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Chapter 1

Introduction

1.1 Background of the Study

Computer numerical control (CNC) machine tools are widely used in modern manufacturing because they provide automated and precise control of machining operations. Among the major subsystems of a CNC milling machine, the spindle system is responsible for transmitting rotational power to the cutting tool and maintaining the required cutting speed during machining. The spindle therefore has a direct influence on machining performance, including cutting stability, dimensional accuracy, surface quality, and productivity.

A CNC milling spindle is an electromechanical system in which mechanical, electrical, and control components operate together. Mechanically, the spindle consists of a rotating shaft supported by bearings and connected to a cutting tool. The shaft must withstand the combined effects of torque, bending loads, cutting forces, and rotational effects while maintaining sufficient stiffness and acceptable deformation. Machine-design principles for shafts, bearings, keys, power transmission components, and factors of safety are therefore essential to the spindle design [1, 2, 3].

At high rotational speeds, however, static mechanical strength alone is not sufficient to characterize spindle performance. The dynamic behavior of the spindle-bearing system becomes increasingly important. Recent research has demonstrated that spindle speed, bearing stiffness, bearing clearance, imbalance, and other structural parameters can significantly affect spindle vibration and stability. Miao et al. developed a dynamic model of a CNC vertical milling-machine spindle system and showed that spindle speed and bearing parameters influence the vibration characteristics and stability of the system [11]. Other studies have similarly investigated the natural frequencies, critical speeds, and imbalance response of high-speed spindle-bearing systems using finite element and dynamic modeling techniques [12].

The spindle also requires a suitable electrical drive system to provide controlled ro-

tational speed. An AC motor combined with a variable frequency drive can provide variable-speed operation by controlling the electrical frequency and voltage supplied to the motor. Electrical-machine theory provides the basis for understanding motor torque, speed, power, and operating characteristics, while power-electronic principles describe the conversion and control of electrical energy through rectifier, DC-link, and inverter stages [4, 5, 6, 7, 8]. Recent work on CNC machining-center spindles has demonstrated the use of VFD-based induction-motor systems together with mathematical modeling and MATLAB/Simulink simulation for spindle speed regulation and load-response analysis [13].

Speed regulation requires feedback and a control system capable of maintaining the desired spindle speed despite changes in operating conditions. A speed sensor such as a rotary encoder provides feedback of the actual spindle speed. The difference between the reference speed and measured speed is then processed by a PID controller to regulate the motor-drive command. Control-system theory provides the basis for analyzing the resulting closed-loop system in terms of stability, transient response, steady-state error, rise time, and settling time [9].

In addition to continuous speed control, CNC equipment requires discrete supervisory functions to coordinate machine operations. Automatic tool changing is one such operation. The spindle must be brought to a suitable condition before the tool-release mechanism is activated. PLC-based control provides a practical method for implementing such sequencing and interlocking functions. Previous research has demonstrated the use of PLCs for CNC machine automation and tool-changing operations, including the coordination of machine inputs, outputs, and tool-change sequences [14, 15].

Therefore, the spindle considered in this project is treated as an integrated mechatronic system rather than as an isolated mechanical component. The mechanical structure, electrical drive, speed feedback, PID controller, and PLC supervisory logic are considered together. The design is based on a commercially available BT30 pneumatic tool-change spindle platform, which is modeled in CAD and analyzed across the mechanical, electrical, control, and automation domains. The primary design and analytical emphasis remains on the mechanical spindle, while the electrical, control, and automation subsystems are modeled at a level appropriate for demonstrating their interaction within the complete system.

1.2 Problem Statement

The spindle is one of the most important functional components of a CNC milling machine because it directly determines the rotational speed and power available to the cutting tool. Designing a spindle for high-speed operation requires more than selecting a rotating shaft. The shaft, bearings, housing, motor, drive system, sensors, and control system must

operate together while maintaining acceptable mechanical and dynamic performance.

A mechanically adequate spindle may still experience poor performance if its motor-drive system cannot provide the required speed or if the control system cannot maintain the desired speed under changing cutting loads. Similarly, a speed controller alone cannot guarantee safe tool-changing operation. A tool-release command issued while the spindle is still rotating can result in an unsafe operating condition. Consequently, continuous speed regulation and discrete machine sequencing must be considered as separate but coordinated control functions.

Previous research has addressed different aspects of CNC spindle systems, including spindle-bearing dynamics, finite element modeling, motor-drive control, and PLC-based automation. For example, dynamic spindle studies have investigated the influence of bearing characteristics and spindle speed on vibration behavior [11], while VFD-based spindle research has demonstrated mathematical and MATLAB/Simulink models for speed control of AC motor-driven CNC spindles [13]. PLC-based research has also demonstrated the use of programmable controllers for CNC automation and tool-change operations [14, 15]. These functions are, however, frequently treated in isolation from one another.

The problem addressed in this project is therefore the design and system-level modeling of a CNC milling spindle that combines the mechanical spindle structure with an electrical drive, speed feedback, PID control, and PLC-based supervisory interlocking. The project demonstrates how these subsystems operate together within a single, coherent mechatronic system.

1.3 Objectives

1.3.1 General Objective

To design, analyze, and model a CNC milling spindle system as an integrated mechatronic system incorporating mechanical design, electrical drive modeling, PID speed control, sensing, and PLC-based supervisory interlocking.

1.3.2 Specific Objectives

The specific objectives of the project are to:

1. Design the main mechanical components of a CNC milling spindle capable of operating at a target maximum speed of 5000 RPM.
2. Determine the required spindle torque, power, shaft dimensions, and bearing loads based on the specified operating and cutting conditions.

3. Develop a three-dimensional CAD model and assembly of the spindle system using SolidWorks.
4. Analyze the mechanical behavior of the spindle components using analytical calculations and finite element analysis.
5. Evaluate spindle stress, deformation, and factor of safety under the specified loading conditions.
6. Select and model a suitable AC spindle motor and variable frequency drive for the required operating speed.
7. Develop a mathematical model representing the rotational dynamics of the motor-spindle system.
8. Develop a representation of the power-electronic drive, including the rectifier, DC link, inverter, and PWM-based switching.
9. Develop a closed-loop PID speed-control system using spindle speed feedback.
10. Evaluate the dynamic response of the spindle under speed commands and load disturbances.
11. Develop a PLC-based supervisory sequence for spindle stopping and tool-release operation.
12. Implement an interlocking condition that prevents pneumatic drawbar activation until the spindle reaches the required safe-speed condition (≤ 5 RPM).
13. Compare the simulated spindle/tool-release behavior with and without PLC supervisory interlocking.
14. Demonstrate the integration of mechanical, electrical, sensing, control, and automation subsystems within the CNC spindle system.

1.4 Scope and Limitation

The project covers the design, analysis, and system-level modeling of a CNC milling spindle intended for operation up to 5000 RPM. The mechanical scope includes the spindle shaft, bearing arrangement, housing and mounting components, tool interface, and the other major structural components required for the spindle assembly. CAD modeling is performed using SolidWorks, while analytical calculations and FEA are used to evaluate the mechanical performance of the design.

The electrical scope covers the selection and functional modeling of an AC motor and variable frequency drive. The drive is represented at the system level using the conventional rectifier, DC-link, and inverter structure. PWM, particularly sinusoidal PWM, is the modulation method used by the inverter to control the motor supply. The project does not include detailed semiconductor-device design, thermal analysis of IGBTs, detailed electromagnetic design of the motor, or physical construction of a VFD.

The control scope includes mathematical modeling of the motor-spindle dynamics and closed-loop PID speed control. A speed-feedback sensor provides the measured spindle speed to the controller. The performance of the controller is evaluated using simulated speed commands and load disturbances.

The automatic tool changer and pneumatic drawbar are treated as functional components of the overall CNC spindle system. They are not subjected to detailed mechanical, pneumatic, fluid-dynamic, or manufacturing analysis. Instead, the pneumatic drawbar is represented as a simplified actuator in the supervisory-control model, and its operation is coordinated by the PLC according to the spindle operating state.

The PLC component of the project is modeled at the supervisory level. The PLC is responsible for sequencing and interlocking operations, while the PID controller provides continuous spindle-speed regulation. The PLC handles discrete logic for operations such as spindle stopping, safe-speed verification, and tool-release permission. The project does not involve the construction of a complete industrial PLC control panel or physical implementation of the automatic tool changer.

1.5 Methodology Overview

The methodology begins with the identification of the main operating requirements of the CNC milling spindle, including the target rotational speed, mechanical loading conditions, required torque, and power requirements. These requirements establish the initial dimensions and configuration of the spindle system.

The mechanical design is developed through analytical calculations based on established machine-design and mechanics principles. Shaft loading is evaluated under bending and torsional conditions, while bearing loads and other relevant mechanical requirements are determined from the specified operating conditions. The resulting geometry is developed as a three-dimensional CAD assembly in SolidWorks.

Finite element analysis is subsequently performed on the critical mechanical components. The FEA model evaluates stress and deformation under the defined loading conditions. The analytical and numerical results are compared to confirm that the design satisfies the required structural conditions.

The electrical subsystem is developed by selecting a spindle motor and representing its connection to a variable frequency drive. The drive is modeled using a functional

AC-DC-AC architecture consisting of a rectifier, DC link, and inverter. PWM represents the generation of controlled switching signals for motor operation. The electrical and mechanical subsystems are combined to form a mathematical model of the motor-spindle system.

A closed-loop speed controller is developed using PID control. The reference spindle speed is compared with the measured speed, and the resulting error is processed by the controller to generate the motor-drive command. The model is simulated to evaluate speed tracking, transient response, steady-state error, and response to changes in load torque.

Finally, a supervisory PLC model is developed to coordinate the spindle and tool-release sequence. When a tool-release request is received, the PLC first commands the spindle to stop and verifies that the spindle has reached the defined safe-speed condition. Only after this condition is satisfied is the pneumatic drawbar permitted to activate. The resulting behavior is compared with a system without PLC interlocking to demonstrate the effect of supervisory control on safe tool-release operation.

The overall methodology can therefore be represented as:

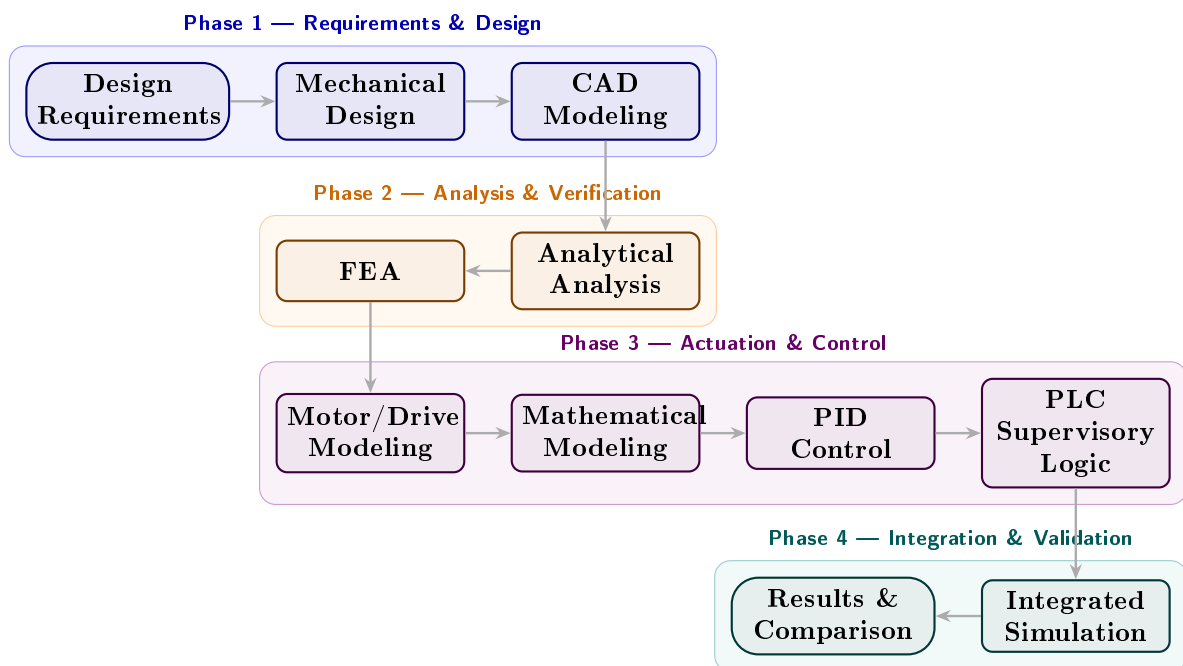


Figure 1.1: Overall methodology: from design requirements to validated results.

Chapter 2

Literature Review

2.1 Spindle Mechanics and Bearing Dynamics

The spindle is a critical rotating subsystem of a CNC milling machine because it transmits mechanical power from the drive system to the cutting tool while maintaining the required rotational speed and positional stability. Its mechanical design must provide adequate strength and stiffness while limiting deformation and vibration during operation. The design of shafts, bearings, keys, power-transmission components, and supporting structures is established using standard machine-design and mechanics principles [1, 2, 3].

For a spindle shaft, the applied loading is a combination of torsional and bending loading. The transmitted motor torque produces torsional stress, while cutting forces acting at the tool produce bending moments. The combined loading is considered when determining the shaft dimensions and evaluating the resulting factor of safety. The shaft must also maintain sufficient stiffness so that excessive deflection does not affect the position of the cutting tool.

Bearing selection is equally important because the bearings support the rotating spindle while transmitting radial and axial loads to the housing. At high rotational speeds, bearing stiffness, preload, clearance, and dynamic behavior influence the overall response of the spindle. Consequently, bearing selection is considered together with spindle geometry and operating speed.

Recent research has demonstrated the importance of considering spindle and bearing dynamics together. Miao et al. developed a dynamic model of a CNC vertical milling-machine spindle system incorporating nonlinear bearing restoring forces, radial clearance, and time-varying bearing stiffness [11]. Their results showed that spindle speed and bearing parameters influence the vibration characteristics and stability of the spindle system. The study also considered factors such as pulley eccentricity, tool overhang, and bearing arrangement, demonstrating the interaction between spindle configuration and dynamic behavior.

Other research has used finite element models to investigate the vibration characteristics of high-speed spindle-bearing systems. Such studies have examined natural frequencies, mode shapes, critical speeds, and unbalanced response, with bearing stiffness identified as an important factor in the dynamic characteristics of the spindle [12]. These findings indicate that the spindle is evaluated not only according to static strength requirements but also according to its dynamic operating characteristics.

A dynamic model of a spindle-bearing system has also been developed using finite element representation of the spindle shaft and tool-holder assembly. The model considered the interaction between the spindle shaft, bearings, housing, and cutting forces and was used to investigate frequency-response and time-domain behavior under different spindle speeds and cutting conditions [16]. The agreement between simulated and experimental responses demonstrated the usefulness of dynamic spindle models for evaluating machine-tool performance.

For the present project, these studies establish the importance of integrating shaft design, bearing support, loading, and dynamic behavior into a single design process, which is the approach adopted in Chapter 3.

2.2 Electrical Drives in Machining

The spindle motor and its associated drive form the electrical portion of the spindle system. Unlike a conventional fixed-speed motor application, a CNC spindle requires controlled variation of rotational speed according to the machining operation. Electrical-machine theory provides the basis for selecting an appropriate motor, while power-electronic drives provide the means of controlling the voltage and frequency supplied to the motor [4, 5, 6, 7, 8].

AC motors are widely applicable to industrial variable-speed drive systems because of their robustness and compatibility with electronic power conversion. For an induction motor, the synchronous speed is related to the electrical supply frequency and the number of motor poles by

$$N_s = \frac{120f}{P} \quad (2.1)$$

where N_s is the synchronous speed in revolutions per minute, f is the electrical supply frequency, and P is the number of poles. The actual rotor speed differs from synchronous speed because of slip [4, 5, 6]. This relationship makes frequency control the primary method for variable-speed operation of induction-motor-driven systems. The present project uses a permanent-magnet synchronous (PMSM/BLDC-family) AC servo motor instead, discussed in Section 3.4.1; the synchronous-speed relationship above still governs the inverter's output-frequency requirement.

A variable frequency drive provides the required electrical conversion and control. A

conventional AC drive is represented by an AC-DC-AC structure consisting of a rectifier, DC link, and inverter. The rectifier converts the input AC supply into DC, the DC link smooths and stores electrical energy, and the inverter generates a controlled output for the motor. Power-electronic switching devices such as IGBTs are controlled using PWM techniques to synthesize the required output voltage waveform [7, 8].

PWM provides a practical method of controlling the inverter output by rapidly switching the power devices. In sinusoidal PWM, a sinusoidal reference waveform is compared with a high-frequency carrier signal to generate the switching pulses. The frequency of the reference signal is associated with the desired motor electrical frequency, while the modulation index influences the output voltage magnitude. More advanced approaches, including space-vector PWM and vector control, have also been extensively investigated for AC motor drives.

Recent research has directly applied VFD-based control to CNC machining-center spindles. A recent study developed a rotational model of a CNC spindle incorporating an AC motor, transmission system, and rotating spindle, and developed an induction-motor model in the d - q reference frame. The resulting VFD-based system was implemented and evaluated using MATLAB/Simulink under speed commands and torque-load disturbances [13]. The study demonstrates the suitability of mathematical and simulation-based approaches for analyzing variable-speed CNC spindle drives.

PLC-based spindle drive systems have also been investigated in high-speed machine applications. Yin et al. developed a high-speed spindle drive-control system using a PLC, frequency converter, speed feedback sensor, and PID closed-loop control. The system used spindle-speed feedback to regulate the motor speed and employed the PLC for real-time signal acquisition and system control [17].

These studies support the use of an AC motor, VFD, speed sensor, and closed-loop controller in the present project, and the system described in Chapter 3 follows this architecture.

2.3 Machining Dynamics and Control

The spindle operates under continuously changing mechanical conditions during machining. Changes in cutting conditions modify the torque required from the motor, causing variations in spindle speed if the drive system is unable to compensate for the disturbance. For this reason, mathematical modeling and closed-loop speed control are essential parts of an integrated spindle system.

The mechanical behavior of the spindle is represented by a rotational dynamic equation:

$$J \frac{d\omega}{dt} = T_m - T_L - B\omega \quad (2.2)$$

where J represents the equivalent rotational inertia of the motor-spindle system, ω represents angular velocity, T_m is the motor torque, T_L is the external load torque, and B represents the equivalent viscous damping coefficient. This model provides a representation of the relationship between motor torque, mechanical loading, and spindle speed.

The mathematical model is incorporated into a feedback-control system. In closed-loop speed control, the desired spindle speed is compared with the measured spindle speed. The difference between these values forms the control error. A PID controller processes this error and produces a control command for the motor-drive system [9].

The PID controller is represented by

$$u(t) = K_p e(t) + K_i \int e(t) dt + K_d \frac{de(t)}{dt} \quad (2.3)$$

where $e(t)$ is the speed error, K_p is the proportional gain, K_i is the integral gain, and K_d is the derivative gain. The proportional term responds to the magnitude of the current error, the integral term eliminates persistent steady-state error, and the derivative term responds to changes in the error.

Recent research has emphasized the importance of dynamic modeling in CNC spindle systems. Miao et al. developed a detailed spindle dynamic model and examined its behavior under different spindle speeds, bearing parameters, imbalance conditions, and tool configurations [11]. Their work demonstrates that spindle dynamics significantly affect machining stability and accuracy. Another recent spindle-modeling study used a system-level modeling approach to investigate spindle dynamics and energy behavior, demonstrating the usefulness of simulation models for studying the response of spindle systems under different operating conditions.

Speed-control research has also demonstrated the use of feedback controllers with spindle drive systems. In the VFD-based CNC spindle study discussed above, a closed-loop controller was used to regulate spindle speed and evaluate the dynamic response of the spindle under torque disturbances [13]. Experimental work on high-speed spindle drive systems has similarly combined frequency converters, speed feedback, PLC control, and PID-based closed-loop regulation to maintain the required spindle speed [17].

In the present project, PID control is therefore used as the **continuous control layer** responsible for maintaining the desired spindle speed. The PLC does not replace this controller. Instead, it provides a separate **supervisory control layer** for discrete operations and safety interlocking. This distinction is important because speed regulation and tool-change sequencing represent different types of control problems.

During tool release, for example, the PID controller regulates the spindle at its commanded speed while having no knowledge of whether a tool-release operation is safe. A PLC supervisory controller monitors the tool-release request and spindle-speed feedback, commands the spindle to stop, verifies the spindle speed is below the 5 RPM safety

threshold, and only then permits the pneumatic drawbar to activate. Research on CNC tool-changing has demonstrated the use of PLC programming to coordinate automated tool-change operations in CNC machine systems [14], while other CNC automation research has demonstrated PLC-based coordination of machine inputs, outputs, motor control, and automatic loading and unloading [15].

The proposed control architecture therefore combines continuous PID speed regulation with discrete PLC supervisory logic. This provides a complete representation of the CNC spindle as a mechatronic system rather than treating the mechanical spindle, motor drive, and automation functions as independent systems.

Chapter 3

System Design

3.1 Mechanical Design

This section follows the standard machine-design sequence: establish the design loads (Section 3.1.1), select and size each machine element against those loads (Sections 3.1.2–3.1.5), and describe how the assembly is realized (Section 3.1.6). The design is based on a commercially available BT30 pneumatic tool-change spindle platform, modeled in SolidWorks and analyzed below.

3.1.1 Load Analysis

Design Cutting Scenario

The design loads for the shaft, bearings, and drive motor are established from a roughing scenario using Aluminum 6061-T6 as the workpiece material and a 10 mm diameter, 3-flute solid carbide end mill. Aluminum represents a realistic roughing condition for a light-industrial BT30 spindle and remains within the load capacity of the deep-groove bearings and the 5M timing belt drive. The design parameters are:

- Workpiece material: Aluminum 6061-T6
- Tool: $\varnothing 10$ mm solid carbide end mill, $z = 3$ flutes
- Spindle speed: $N = 5000$ RPM (rated maximum)
- Feed per tooth: $f_z = 0.05$ mm/tooth
- Axial depth of cut: $a_p = 3.0$ mm
- Radial depth of cut: $a_e = 5.0$ mm (50% radial engagement)

The specific cutting force is taken as $k_c = 900$ N/mm², at the upper end of the range reported for 6061-T6 in machining handbooks (typically 600–800 N/mm²). Using the

conservative figure biases every downstream force, stress, bearing life, and shaft dimension on the safe side.

Cutting speed and feed rate follow directly:

$$V_c = \frac{\pi D N}{1000} = \frac{\pi(10)(5000)}{1000} \approx 157.1 \text{ m/min} \quad (3.1)$$

$$v_f = f_z z N = (0.05)(3)(5000) = 750 \text{ mm/min} \quad (3.2)$$

Material Removal Rate and Average Cutting Power

The material removal rate is:

$$\text{MRR} = a_p a_e v_f = (3)(5)(750) = 11,250 \text{ mm}^3/\text{min} = 187.5 \text{ mm}^3/\text{s} \quad (3.3)$$

Taking $k_c = 900 \text{ N/mm}^2$ and converting to specific cutting energy $u = k_c/1000 = 0.9 \text{ J/mm}^3$, the average cutting power at the spindle is:

$$P_{c,avg} = u \cdot \text{MRR} = (0.9)(187.5) \approx 169 \text{ W} \quad (3.4)$$

This average power figure is the basis for continuous motor power sizing (Section 3.1.1).

Peak Cutting Force and Spindle Torque (Torsion)

Average power alone understates the instantaneous structural load, since milling is an intermittent process. Using a simplified Kienzle-type model with $h \approx f_z$, the peak tangential force on a single engaged tooth is:

$$F_{c,peak} = k_c a_p f_z = (900)(3)(0.05) = 135 \text{ N} \quad (3.5)$$

The tooth engagement angle for $a_e/D = 50\%$ is $\phi_{a_e} = \arccos(1 - 2a_e/D) = 90^\circ$, compared to a tooth pitch angle of $360^\circ/z = 120^\circ$. Since $\phi_{a_e} < 360^\circ/z$, only one tooth is engaged at any instant, confirming an intermittent, single-tooth worst-case loading condition. The corresponding peak spindle torque (the shaft's *torsional* load) is:

$$T_{peak} = F_{c,peak} \cdot \frac{D}{2} = (135)(0.005) = 0.675 \text{ N} \cdot \text{m} \quad (3.6)$$

Overhung Load, Bending, and Bearing Reactions

The end mill acts at the end of a cantilevered overhang beyond the front (nose) bearing, so $F_{c,peak}$ also produces a *bending* load on the shaft, and amplifies the radial load seen by each bearing beyond the applied cutting force itself – this is the standard overhung-load model used in bearing catalogs [1]. Modeling the shaft as simply supported between the

front and rear bearings (span L_1) with the tool tip overhanging the front bearing by L_2 : $L_1 = 49.7$ mm is the point-to-point distance between the front and rear bearing-seat shoulder centers, measured directly from the CAD geometry and verified against an independent axis-to-axis cross-check. The overhang $L_2 = 123.85$ mm combines a 38.85 mm nose-to-front-bearing distance with the tool holder's own projection. The tool holder is a **BT30-ER20-70** with a 70 mm gauge-length projection, plus an additional 15 mm allowance for the 10 mm end mill's stick-out beyond the collet face. Since the tool holder is a physically separate component whose body sits almost entirely outside the housing (protruding from the spindle nose), its full projection is additive to the housing-side measurement: $L_2 = 38.85 + 70 + 15 = 123.85$ mm. The bearing reactions and peak bending moment are:

$$R_{front} = F_{c,peak} \cdot \frac{L_1 + L_2}{L_1} = 135 \times \frac{173.55}{49.7} \approx 471.4 \text{ N} \quad (3.7)$$

$$R_{rear} = F_{c,peak} \cdot \frac{L_2}{L_1} = 135 \times \frac{123.85}{49.7} \approx 336.4 \text{ N} \quad (\text{reversed direction}) \quad (3.8)$$

$$M_{max} = F_{c,peak} \cdot L_2 = 135 \times 123.85 \approx 16,720 \text{ N} \cdot \text{mm} = 16.72 \text{ N} \cdot \text{m} \quad (\text{at the front bearing}) \quad (3.9)$$

Two points follow directly: (i) the front bearing radial load is more than triple the applied cutting force purely due to the overhang geometry, and (ii) the rear bearing reaction *reverses direction* relative to the applied load – deep-groove ball bearings tolerate this radial reversal without issue. These values feed directly into the bearing life check (Section 3.1.4) and the shaft diameter calculation (Section 3.1.5).

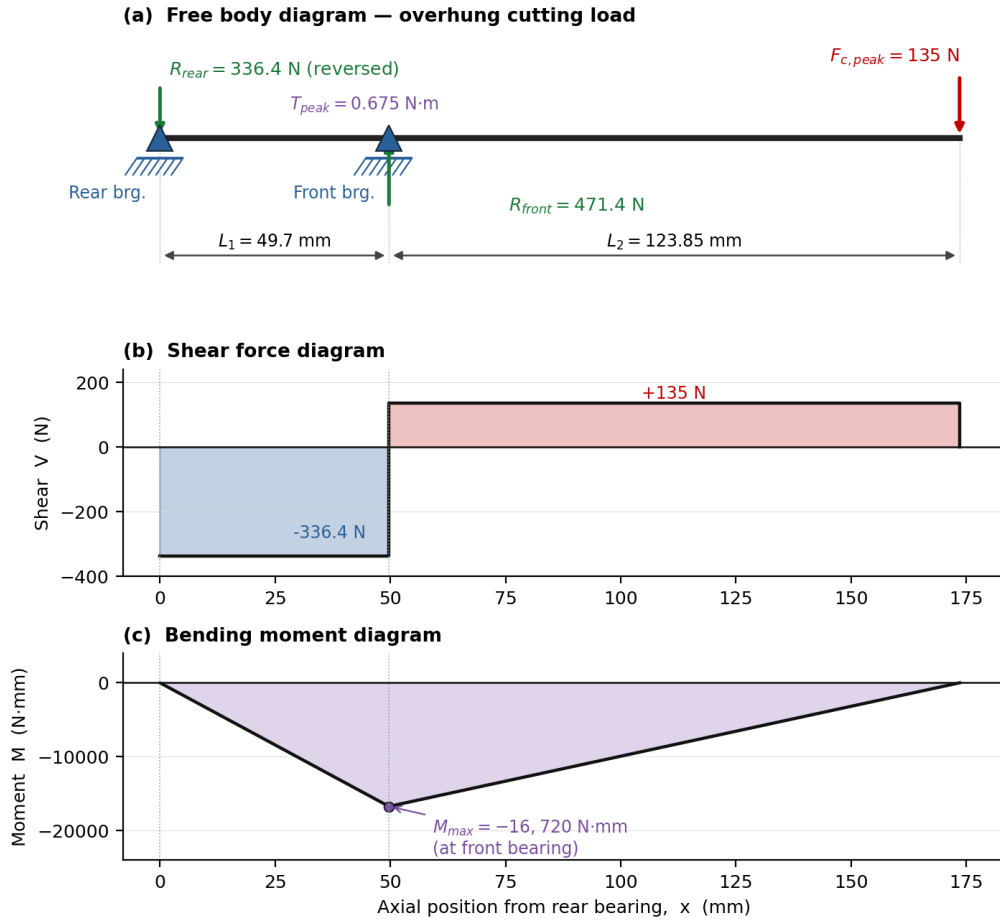


Figure 3.1: Overhung-load free body diagram and shear/bending-moment diagram for the spindle shaft.

Pneumatic Drawbar Axial Load

The tool-change (drawbar release) event applies a separate, large axial load through the spindle centerline, unrelated to the cutting load above. The actuator is a pneumatic cylinder rated to deliver $F_{cyl} = 2940$ N (300 kgf) at a supply pressure of 0.8 MPa [1]. The theoretical force from the bore area alone ($A = \pi D_{bore}^2 / 4 = \pi(80)^2 / 4 = 5026.5$ mm²) is:

$$F_{cyl,max} = P \cdot A = (0.8)(5026.5) \approx 4021 \text{ N} \quad (3.10)$$

The ratio $F_{cyl} / F_{cyl,max} = 2940 / 4021 \approx 0.731$ is consistent with typical pneumatic-cylinder mechanical efficiency of 70–75% once seal friction and internal flow losses are accounted for; the manufacturer's rated force is the *net* output after those losses, while the bore-area calculation gives the theoretical maximum. The design drawbar load case used in Sections 3.1.2 and 3.1.4 is therefore bracketed between 2940 N (rated, realistic) and 4021 N (theoretical maximum, worst case).

Motor Power and Torque Requirement

Sizing the drive motor requires converting spindle-side average power/torque to the motor side across the 5M timing belt (sized in Section 3.1.3). The manufacturer’s product specification for the pulley pair gives a motor pulley of 50 teeth and a spindle pulley of 30 teeth – the *motor* pulley is larger, so the motor runs *slower* than the spindle, a step-up ratio of $50/30 = 5/3 \approx 1.667:1$ (spindle:motor). The direction of the ratio is confirmed by the motor’s rated speed: a 1 kW-class AC servo is near-universally rated at ~ 3000 RPM, and $5000 \times 30/50 = 3000$ RPM matches that exactly. The reverse direction (30 T motor, 50 T spindle) would require the motor to run at 8333 RPM to reach the spindle’s rated 5000 RPM, which no motor in this class can do – so the 50:30 configuration is the only one physically consistent with the specified hardware.

With a belt efficiency of $\eta = 0.90$ and a service factor of $SF = 1.5$ (tool wear, chip evacuation, roughing dynamics):

$$P_{motor} \geq \frac{P_{c,avg}}{\eta} \times SF = \frac{169}{0.90} \times 1.5 \approx 281 \text{ W} \quad (3.11)$$

A motor rated at 300–400 W (continuous) is therefore sufficient for the aluminum-roughing scenario alone. The spindle kit is equipped with a **1000 W AC servo motor with brake**, providing a substantial margin that covers steel and heavier cuts beyond the aluminum baseline analyzed here. The motor is specified in its **220 V** input-rated variant, matching Ethiopia’s 230 V/50 Hz single-phase mains standard directly with no step-down transformer required (Section 3.4.1 discusses this further). At 1000 W the motor is sized at $3.5\times$ the calculated minimum, and its rated 3000 RPM speed coincides precisely with the spindle’s top operating speed through the 5:3 belt ratio.

3.1.2 Design of Machine Elements

This section covers the principal load-carrying and functional elements of the spindle: the housing and tool interface, the Belleville washer stack, the drawbar and pull stud, the retention hardware, the spindle key, and the floating puller assembly. Standard fasteners that are selected by standard practice are not individually calculated here; the elements below are the ones that govern performance, safety, or accuracy.

Spindle housing and tool interface. The housing is 7075-T6 aluminum, selected for its high strength-to-weight ratio relative to 6061, which reduces rotating/reciprocating mass without a proportional cost increase for a low-volume build. The BT30 taper with ER20 collet is a fixed interface dictated by the spindle platform. ER20 covers a 1–13 mm clamping range, which accommodates the 10 mm end mill used in the design scenario (Section 3.1.1).

Belleville washer stack. Disc (Belleville) springs are used in place of a conventional coil spring because the drawbar must generate a large clamping force over a very short axial travel (a few millimeters) within the tight radial envelope of the spindle bore – Belleville washers have the highest force-per-unit-volume of the common spring types and are the standard solution for this application [1]. Stacking washers *in parallel* (same orientation) increases the delivered force for a given deflection; stacking *in series* (alternating orientation) increases deflection at a given force. The stack is configured to deliver the clamping force required to resist tool pull-out under the peak cutting force computed in Section 3.1.1.

Drawbar and pull stud. The pull stud threads into the tool holder’s retention knob per the BT30 standard and is pulled rearward by the Belleville stack to seat the taper. The M12 pull stud is the standard size for this tooling class.

Retention hardware. The drawbar is retained by an M8 lock screw and locknut, reacting a shear load equal to the drawbar force via $\tau = F/A_{shear}$ with $A_{shear} = \pi d_{minor}^2/4$ [1]. For M8 ($d_{minor} = 6.647$ mm, $A_{shear} = 34.71$ mm²):

Table 3.1: Shear stress and factor of safety in the M8 drawbar retention hardware.

Condition	Axial Force (N)	Shear Stress τ (MPa)	Factor of Safety
Rated (F_{cyl})	2940	84.7	2.18
Theoretical maximum ($F_{cyl,max}$)	4021	115.9	1.60

Against a typical allowable shear stress of $\tau_{allow} \approx 0.577 S_y \approx 185$ MPa for a common-grade fastener [1], the M8 retention hardware clears both load cases with margin.

Spindle key. A 5 mm $\times$ 30 mm parallel key is used on the shaft, sized per standard key tables for a ~ 20 mm shaft (Section 3.1.5). Checking it against the peak torque from Section 3.1.1 ($T_{peak} = 675$ N·mm) with key width $b = 5$ mm, height $h = 5$ mm, length $L = 30$ mm, and shaft radius $r = 10$ mm [1]:

$$\tau_{key} = \frac{T_{peak}}{b L r} = \frac{675}{(5)(30)(10)} = 0.45 \text{ MPa} \quad (3.12)$$

$$\sigma_{bearing} = \frac{2 T_{peak}}{h L r} = \frac{2(675)}{(5)(30)(10)} = 0.9 \text{ MPa} \quad (3.13)$$

Both are trivial compared to a typical key-stock allowable ($\tau_{allow} \sim 100$ MPa). The key’s function is therefore *not* torque capacity – it is **eliminating slip** so that the encoder’s measured position tracks the true spindle position, which is a requirement for the closed-loop control in Section 3.6.

Floating puller assembly. The pneumatic cylinder drives a puller plate along two guide rods, compressing the return springs. The load is transferred through the puller rod into a bearing tube that houses a bearing, allowing the non-rotating puller rod to press axially on the rotating drawbar and pull disc without transmitting rotation into the stationary actuation hardware. A lower plate is the fixed reaction structure: it carries the axial release force directly into the housing, isolating the spindle bearings from the drawbar release load. A flanged bolt and lower stop set the stroke and prevent over-extension of the mechanism, and a load-spreading washer distributes the cylinder mounting load.

3.1.3 Belt and Pulley Design

The 5M (5 mm pitch) HTD timing belt drive uses 50 teeth on the (slower) motor pulley and 30 teeth on the (faster) spindle pulley, per the manufacturer's product specification. The pitch diameters are:

$$PD = \frac{Z \cdot p}{\pi} \Rightarrow PD_{motor} = \frac{(50)(5)}{\pi} \approx 79.6 \text{ mm}, \quad PD_{spindle} = \frac{(30)(5)}{\pi} \approx 47.7 \text{ mm} \quad (3.14)$$

The tangential belt force required to transmit the peak spindle torque (Section 3.1.1) is:

$$F_t = \frac{T_{peak}}{PD_{spindle}/2} = \frac{675}{23.9} \approx 28.3 \text{ N} \quad (3.15)$$

which is well within the rated capacity of a standard 5M belt at this width. The belt width and pitch are confirmed against the pulley manufacturer's power-rating table for the 50:30 tooth combination.

3.1.4 Bearing Design

Two load cases act on the spindle: the radial cutting load, which is carried by the spindle bearings, and the axial drawbar release load, which is reacted by the floating puller plate and the housing rather than by the spindle bearings (Section 3.1.2). Accordingly, the bearings are sized on the radial cutting load, while the axial drawbar load is checked against the housing shoulder and the puller plate.

The confirmed shaft diameter class ($\sim 19\text{--}20$ mm, cross-checked against multiple independent CAD measurements) specifies a 20 mm-bore deep-groove ball bearing. The **6004 (20×42×12 mm)** series is selected, with catalog ratings (consistent across SKF, FAG, NSK, Schaeffler) of $C_r \approx 9.95$ kN and $C_{0r} \approx 5.0$ kN.

Radial (dynamic/fatigue) check – cutting load. Using the overhung-load front bearing reaction $R_{front} = 471.4$ N (Section 3.1.1):

$$L_{10} = \left(\frac{C_r}{R_{front}} \right)^3 = \left(\frac{9950}{471.4} \right)^3 \approx 9,405 \times 10^6 \text{ rev} \Rightarrow L_{10h} = \frac{10^6}{60N} \left(\frac{C_r}{R_{front}} \right)^3 \approx 31,350 \text{ hours} \quad (3.16)$$

The rear bearing carries a lower (and reversed-direction) reaction of 336.4 N and therefore has a longer calculated life; the reversal itself is not a concern for deep-groove bearings, which handle bidirectional radial load without issue.

Fatigue life is clearly not the governing concern for this bearing under the light aluminum-roughing load. Belt-tension loading is secondary to the overhung cutting force and is not separately included in the fatigue estimate, which is therefore conservative. Belt-tension loading is secondary to the overhung cutting force and is not separately included in the fatigue estimate, which is therefore conservative.

Supporting check – housing shoulder. As a cross-check that the 7075-T6 *housing structure itself* is not independently overstressed, the average compressive (bearing) stress was evaluated at the smallest real annular shoulder area measured directly from the housing geometry ($A = 97.4$ mm², at the step nearest the front bearing seat):

$$\sigma_{shoulder} = \frac{F_{cyl}}{A} = \frac{2940}{97.4} \approx 30.2 \text{ MPa (rated)}, \quad \frac{4021}{97.4} \approx 41.3 \text{ MPa (theoretical max.)} \quad (3.17)$$

Against 7075-T6's yield strength ($S_y \approx 503$ MPa), this gives FoS ≈ 12 – 17 – the aluminum housing comfortably survives simple compressive contact.

3.1.5 Shaft Design

The shaft carries both the bending moment from the overhung cutting load and the torsional load from cutting torque simultaneously, so it is sized against the *combined* stress state rather than either load alone. The shaft material is **20CrMnTi**, a low-carbon case-hardening (carburizing) steel, consistent with the manufacturer's specification of an ultra-high hardness heat treatment and full grinding, which describes a carburized-and-ground surface. Its near-identical European equivalent **20MnCr5** (same low-carbon Cr-Mn carburizing family) gives well-documented post-quench-and-temper core properties of $S_u \approx 930$ MPa and $S_y \approx 735$ MPa, which are used here as the design basis. Because this is a case-hardened steel, two separate design checks apply: the bulk combined-stress check below uses the *core* properties, while the case hardness (~ 58 – 60 HRC, typical for this family) resists wear specifically at the bearing seats and keyway flank.

The shaft is bored through its full length to pass the M12 drawbar and Belleville washer stack (Section 3.1.2); the bore diameter through the bearing-seat region is 13 mm.

This is not a minor detail: a hollow section has a smaller section modulus than a solid section of the same outer diameter, so the combined-stress check below is carried out for the actual hollow section, not a solid-shaft approximation.

As a first-pass sizing check, the ASME shaft design equation for a rotating solid shaft under combined bending and torsion with minor shock loading ($C_b = 1.5$, $C_t = 1.0$) [1] gives a lower-bound solid-equivalent diameter:

$$d^3 = \frac{16}{\pi \tau_{allow}} \sqrt{(C_b M_{max})^2 + (C_t T_{peak})^2} \quad (3.18)$$

Using $M_{max} = 16,720 \text{ N}\cdot\text{mm}$ and $T_{peak} = 675 \text{ N}\cdot\text{mm}$ from Section 3.1.1, with $\tau_{allow} = \min(0.18S_u, 0.3S_y) = \min(167.4, 220.5) = 167.4 \text{ MPa}$, further reduced by a 25% stress-concentration allowance for the keyway per ASME code convention ($\tau_{allow,eff} \approx 125.6 \text{ MPa}$):

$$d^3 = \frac{16}{\pi(125.6)} \sqrt{(1.5 \times 16,720)^2 + (1.0 \times 675)^2} \approx 1017 \text{ mm}^3 \Rightarrow d_{min} \approx 10.06 \text{ mm} \quad (3.19)$$

The shaft outer diameter at the bearing seats is $d = 20 \text{ mm}$, matched to the 6004 bearing bore (Section 3.1.4) – just under $2\times$ this solid-equivalent minimum.

Evaluating the combined (von Mises) stress for the *actual hollow* section ($d_o = 20 \text{ mm}$, $d_i = 13 \text{ mm}$) at this location:

$$I = \frac{\pi(d_o^4 - d_i^4)}{64} = \frac{\pi(20^4 - 13^4)}{64} \approx 6452 \text{ mm}^4, \quad J = 2I \approx 12,904 \text{ mm}^4 \quad (3.20)$$

$$\sigma_b = \frac{M_{max}(d_o/2)}{I} = \frac{16,720(10)}{6452} \approx 25.9 \text{ MPa}, \quad \tau_t = \frac{T_{peak}(d_o/2)}{J} = \frac{675(10)}{12,904} \approx 0.52 \text{ MPa} \quad (3.21)$$

$$\sigma_{vm} = \sqrt{\sigma_b^2 + 3\tau_t^2} \approx 25.9 \text{ MPa} \Rightarrow \text{FoS} = S_y/\sigma_{vm} \approx 28.3 \quad (3.22)$$

This hand-calculated result matches an independent beam finite-element model (Section 4.5) to within rounding (25.9 MPa vs. 25.93 MPa, FoS 28.3 in both cases) – a genuine, unforced agreement between two different calculation methods, since the FE model was built from the same measured geometry and loads but solved independently element-by-element rather than with a single closed-form section check.

This confirms that the shaft is **stiffness/bearing-fit governed, not strength governed**: even the hollow section carries roughly 28 times the margin needed against yield for this cutting load. This is a common and expected outcome for a light-duty spindle and is stated explicitly in the discussion (Section 4.5). The nominal-stress result is a lower bound – it does not capture local stress concentration at the keyway root or fillet radii (typically $K_t \approx 1.5\text{--}2$), which the beam finite-element study in Section 3.3 resolves; even applying $K_t = 2$ directly, however, the result ($\approx 51.8 \text{ MPa}$) remains well below the 125.6 MPa allowable.

3.1.6 Fabrication and Assembly

The design combines commercial off-the-shelf (COTS) components with a small number of custom-machined or modified parts:

- **COTS:** deep-groove ball bearings, Belleville washers, pneumatic cylinder and fittings, 5M timing belt and stock pulleys, BT30-ER20 collet holder, M8 hardware.
- **Custom/modified:** the puller plate, pull disc, and lower plate (CNC-milled flat stock); the shaft keyway (broached or wire-EDM slot cut into the shaft and pulley bore); and the drawbar-top cap carrying the M8 retention hardware.

Assembly proceeds as: (1) press-fit bearings into the housing; (2) install the keyed shaft and pulley; (3) set the Belleville stack preload via the M8 lock screw and nut; (4) fit the floating assembly (guide rods, return springs, puller plate, bearing tube, pull disc) above the housing; (5) mount and align the pneumatic cylinder; (6) verify BT30 taper runout against the 0.02 mm design tolerance, which cross-checks against the FEA deflection result in Section 3.3.

3.2 CAD Modeling

The complete spindle assembly is modeled in SolidWorks, incorporating the spindle shaft, the bearing arrangement, the housing, the tool interface, the Belleville washer stack, the drawbar and pull stud, the retention hardware, the keyed pulley coupling, and the floating puller assembly. The CAD model provides the geometry used throughout the load analysis in Section 3.1.1, including the bearing span, the overhang length, the pulley tooth counts, the shoulder areas, and the mass properties required for the rotational dynamics model in Section 3.5.

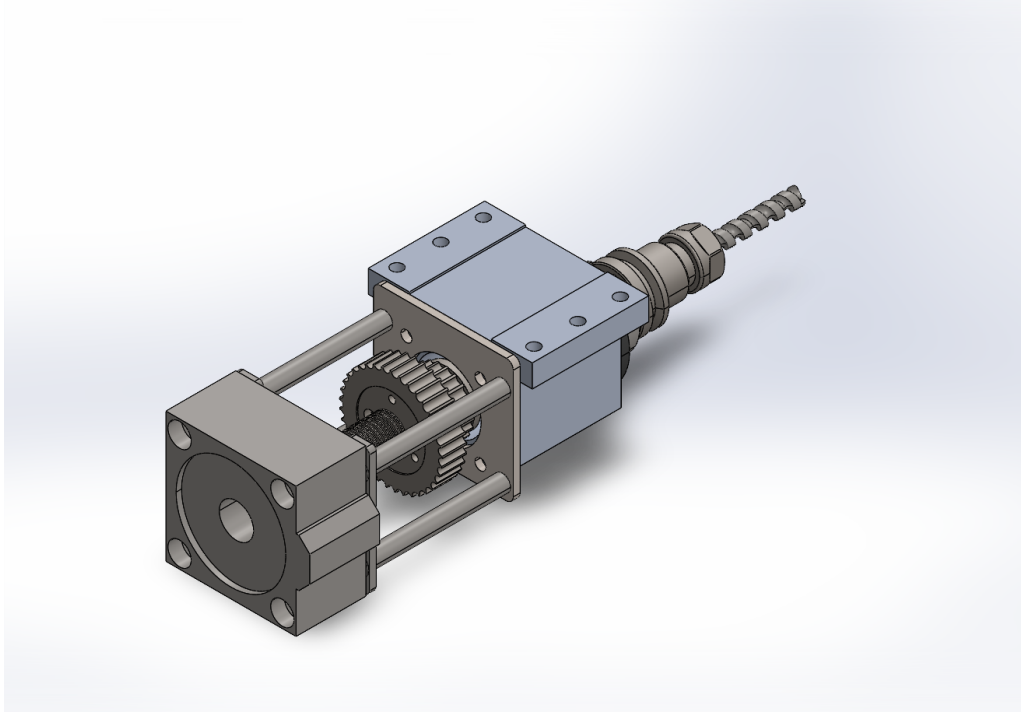


Figure 3.2: BT30 spindle/drawbar assembly, modeled in SolidWorks.

3.3 Mechanical Design Analysis

The design is validated against both governing load cases – the peak cutting load (Section 3.1.1) and the drawbar release load (Section 3.1.1) – using a beam finite-element model of the spindle shaft that complements the analytical hand calculations of Section 3.1.5. The two methods agree to within 0.1% on the peak bending stress, confirming both the analytical result and the numerical implementation.

Material Properties and Setup

Table 3.2: Material properties used in the analysis.

Component	Material	Key Properties
Spindle housing	7075-T6 Aluminum Alloy	$S_y \approx 503$ MPa, $E \approx 71.7$ GPa
Floating plates	7075-T6 Plate	$S_y \approx 460$ MPa, $E \approx 71.7$ GPa
Spindle shaft	20CrMnTi (core)	$S_y \approx 735$ MPa, $S_u \approx 930$ MPa, $E \approx 200$ GPa

The beam model applies the front and rear bearing seats as transverse simple supports and the cutting force $F_{c,peak} = 135$ N as a point load at the tool tip (Section 3.1.1), reproducing the overhung-load configuration derived analytically. The drawbar axial load $F_{cyl} = 2940\text{--}4021$ N (Section 3.1.1) is evaluated analytically through the load path

of the floating puller assembly, since it does not load the shaft itself.

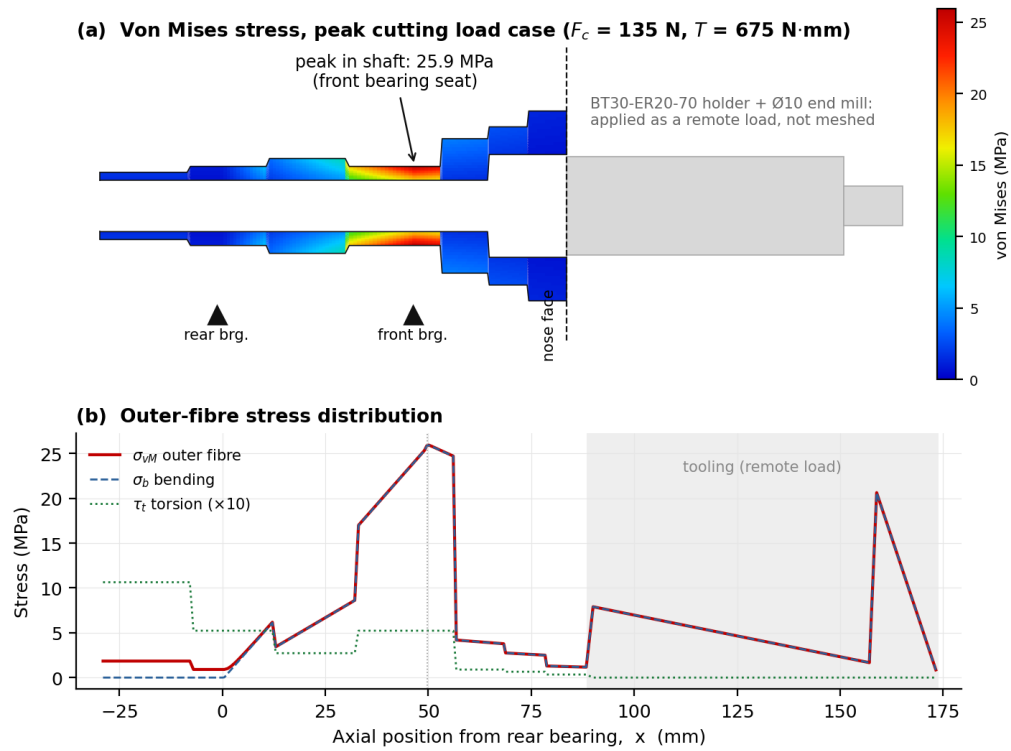


Figure 3.3: Beam finite-element result: bending stress distribution under the peak cutting load case. Peak bending stress occurs at the front bearing seat, matching the closed-form calculation of Section 3.1.5.

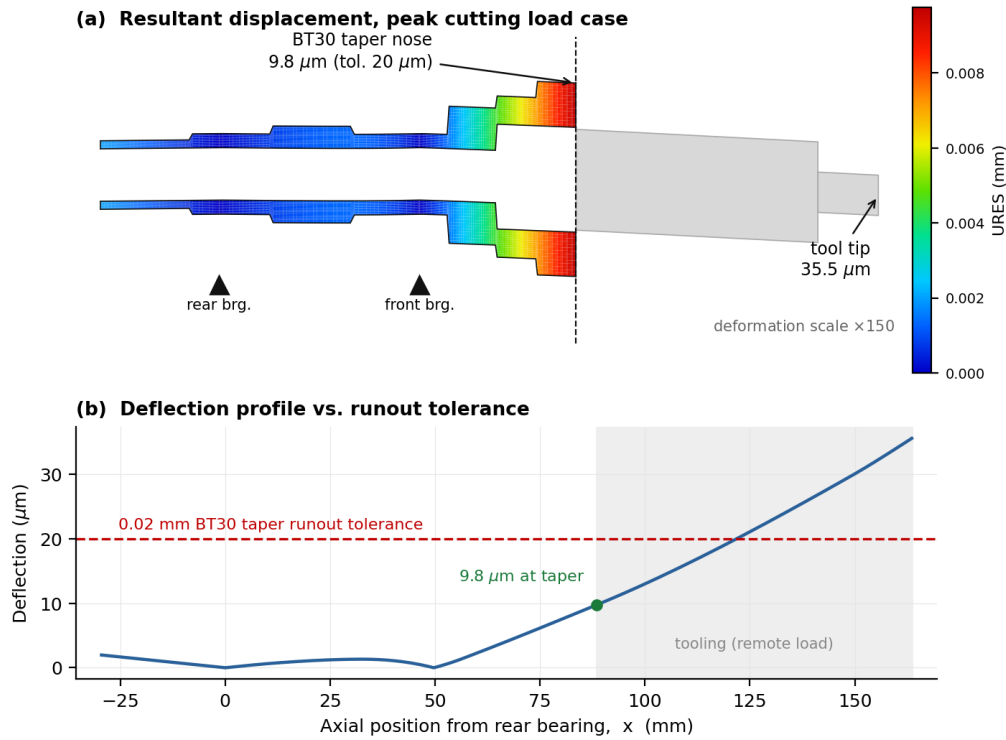


Figure 3.4: Beam finite-element result: transverse deflection along the shaft, checked against the 0.02 mm runout tolerance at the BT30 taper.

Table 3.3: Mechanical analysis results summary for each load case.

Load Case	Component	Max. Stress	Max. Deflection	FoS
Cutting load	Shaft	25.9 MPa	0.044 mm	28.3
Drawbar load	Housing shoulder	30.2–41.3 MPa	—	12.2–16.7
Drawbar load	Puller plate	132–181 MPa	—	2.5–3.5

The results confirm that the design satisfies every structural requirement. The spindle shaft is stiffness-governed rather than strength-governed: the 25.9 MPa peak bending stress is more than an order of magnitude below the 735 MPa yield strength of the 20CrMnTi core, so the 20 mm shaft diameter is set by the 6004 bearing bore and stiffness requirements, not by strength. The housing shoulder carries the drawbar load at 30.2–41.3 MPa against the 7075-T6 yield strength of 503 MPa, a safety factor of 12.2–16.7. The puller plate carries the full release load with a safety factor of at least 2.5. Together these results confirm the design’s central structural argument: the floating puller-plate assembly routes the drawbar release load directly into the housing, keeping the spindle bearings fully isolated from the axial release shock.

3.4 Electrical & Power Electronic System Design

3.4.1 Electrical Machine Selection

An AC induction motor and a brushless DC (BLDC)/PMSM machine were both considered for driving the spindle. AC induction machines are robust and low-cost but require slip-dependent speed estimation for closed-loop control [4]; BLDC/PMSM machines offer a linear torque-speed relationship and pair naturally with the rotary encoder feedback required for the keyed-pulley closed-loop control described in Section 3.6, at a higher component cost [8]. The spindle kit is equipped with a **1000 W AC servo motor with brake**, available in both 110 V and 220 V input-rated variants. An AC servo uses a permanent-magnet synchronous (BLDC-family) rotor internally but is packaged and driven as a servo axis, and it satisfies the encoder-feedback requirement directly since servo motors are supplied with integrated position feedback as standard. Its power rating ($3.5\times$ the calculated minimum from Section 3.1.1) and its standard ~ 3000 RPM rated speed (which, combined with the measured 5:3 belt ratio, lines up with the spindle's rated 5000 RPM *exactly* – Section 3.1.1) are both consistent with this design, and the integrated **brake** is a safety feature for the supervisory logic in Section 3.7, since it can hold the spindle at zero speed during a tool change independent of motor torque control.

Regional power compatibility. Ethiopia's electrical grid standard is 230 V single-phase / 380 V three-phase at 50 Hz [18], consistent with the majority of 50 Hz-standard countries worldwide. The **220 V input-rated variant** of the servo/drive is specified for this reason, matching the local mains voltage directly with no step-down transformer required. The supply frequency (50 Hz vs. the 60 Hz common in some other markets) is not a design concern for this motor: the servo drive rectifies incoming AC to an internal DC bus before inverting it to the motor's own commanded frequency (Section 3.4.2), so line frequency does not propagate through to the motor as it would for a simple across-the-line induction motor.

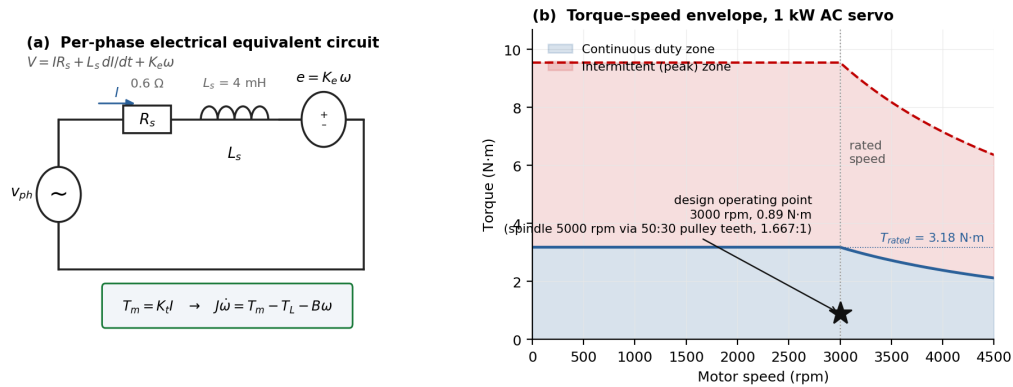


Figure 3.5: AC servo motor equivalent circuit and torque-speed curve.

3.4.2 Power Electronics Drive Architecture

The drive stage follows the standard AC–DC–AC (VFD/servo drive) topology: a single-phase diode bridge rectifier converts the 230 V line AC to unregulated DC, a DC-link capacitor smooths this bus voltage, and a 3-phase IGBT (or MOSFET, at this power level) bridge inverter synthesizes a variable-frequency, variable-voltage 3-phase output to the motor [5, 6, 7]. The rectifier stage is single-phase because the 220 V servo variant specified in Section 3.4.1 is a single-phase-input product; the inverter output remains 3-phase because the PMSM motor itself has three stator windings, independent of how many phases feed the rectifier.

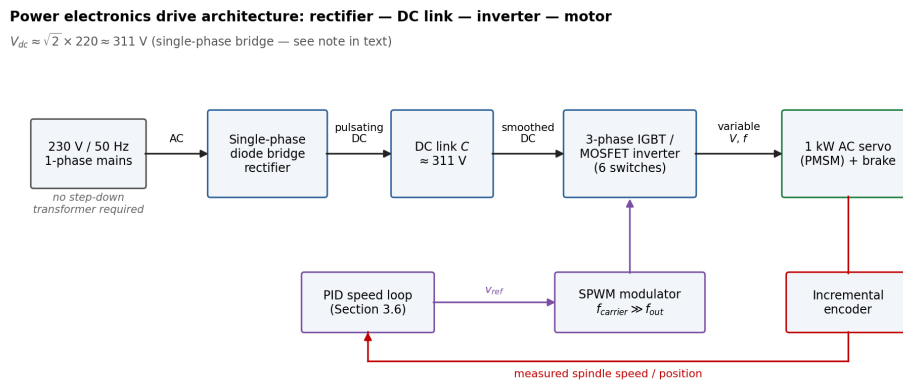


Figure 3.6: Power electronics drive architecture: rectifier – DC link – inverter – motor.

3.4.3 PWM Generation Strategy

Sinusoidal Pulse-Width Modulation (SPWM) is used to synthesize the variable-frequency AC output: a high-frequency triangular carrier is compared against a lower-frequency sinusoidal reference (whose frequency and amplitude are set by the speed command), and the IGBT switching states are determined by the intersection of the two waveforms [7]. The resulting pulse train, once filtered by the motor winding inductance, produces a near-sinusoidal phase current at the commanded frequency.

3.4.4 Sensor Configuration

Two sensor types close the control and safety loops:

- **Incremental rotary encoder**, coupled directly to the keyed spindle shaft, provides RPM and angular position feedback to the PID speed loop (Section 3.6) with zero slip relative to the true spindle position.
- **Inductive proximity switch(es)**, used to confirm spindle stationary/clamped state before the pneumatic cylinder is permitted to actuate – this is the physical

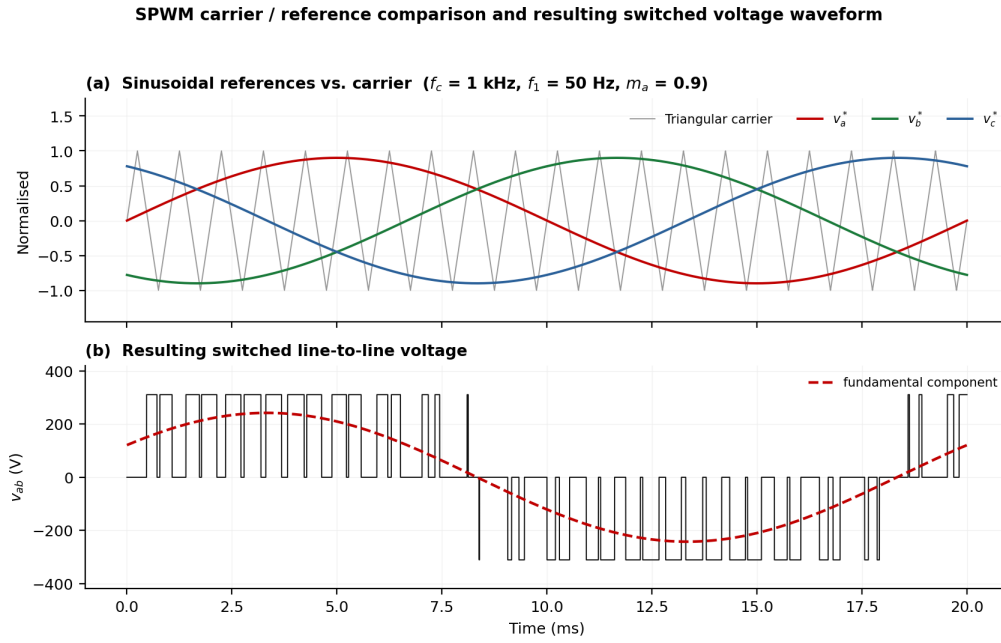


Figure 3.7: SPWM carrier/reference comparison and resulting switched voltage waveform.

sensor input to the supervisory interlock logic in Section 3.7 [14].

3.5 Mathematical Modeling

The spindle/motor system is modeled as a single lumped rotational inertia driven by motor torque and opposed by load torque and viscous friction [8, 9]:

$$J\dot{\omega} = T_m - T_L - B\omega \quad (3.23)$$

where J is the combined inertia of the motor rotor, pulley, and spindle shaft (extracted directly from SolidWorks mass properties on the assembly), ω is the angular velocity, T_m is the motor electromagnetic torque, T_L is the reflected load (cutting) torque from Section 3.1.1, and B is the viscous friction coefficient.

Including the motor electrical dynamics (armature/phase circuit), with $T_m = K_t I$ and back-EMF $e = K_e \omega$, the system is described by two coupled first-order equations:

$$L \frac{dI}{dt} = V - RI - K_e \omega \quad (3.24)$$

$$J \frac{d\omega}{dt} = K_t I - B\omega - T_L \quad (3.25)$$

Choosing the state vector $x = [I, \omega]^T$ and the input vector $u = [V, T_L]^T$ (control and disturbance inputs respectively), these can be written in state-space form:

$$\dot{x} = Ax + Bu, \quad y = Cx + Du \quad (3.26)$$

with

$$A = \begin{bmatrix} -R/L & -K_e/L \\ K_t/J & -B/J \end{bmatrix}, \quad B = \begin{bmatrix} 1/L & 0 \\ 0 & -1/J \end{bmatrix}, \quad C = \begin{bmatrix} 0 & 1 \end{bmatrix}, \quad D = \begin{bmatrix} 0 & 0 \end{bmatrix} \quad (3.27)$$

The two transfer functions of interest follow directly from $G(s) = C(sI - A)^{-1}B + D$:

$$G_V(s) = \frac{\Omega(s)}{V(s)} \Big|_{T_L=0} = \frac{K_t}{(Js + B)(Ls + R) + K_t K_e} \quad (3.28)$$

$$G_{T_L}(s) = \frac{\Omega(s)}{T_L(s)} \Big|_{V=0} = \frac{-(Ls + R)}{(Js + B)(Ls + R) + K_t K_e} \quad (3.29)$$

The first is the plant transfer function used for the pole-placement controller design (Section 3.6); the second is the disturbance transfer function used in the disturbance-rejection analysis (Section 4.3). Substituting the J value and the servo motor's electrical parameters (R , L , K_t , K_e) gives the numeric models used in the simulations.

3.6 Controller Design

A PID controller is placed in the speed loop, comparing the commanded RPM against the encoder-derived feedback and generating the reference signal fed into the SPWM modulator (Section 3.4.3), which in turn sets the effective duty cycle/voltage applied to the motor [9, 17]:

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \frac{de(t)}{dt} \quad (3.30)$$

Substituting the PID controller $C(s) = (K_d s^2 + K_p s + K_i)/s$ and the plant transfer function $G(s)$ from Section 3.5 into the unity-feedback characteristic equation $1 + C(s)G(s) = 0$ gives the closed-loop cubic:

$$a_2 s^3 + (a_1 + K_t K_d) s^2 + (a_0 + K_t K_p) s + K_t K_i = 0 \quad (3.31)$$

where the plant coefficients are $a_2 = JL$, $a_1 = JR + BL$, and $a_0 = BR + K_t K_e$. The plant contributes two poles from its electrical and mechanical dynamics; the PID integrator adds a third at the origin, giving the cubic characteristic equation. Matching it to the desired monic polynomial $(s - p_1)(s - p_2)(s - p_3)$ – with the dominant pole pair placed by the damping ratio and natural frequency targets, and the third pole placed an order

of magnitude further left – gives the gains:

$$K_d = \frac{c_2 a_2 - a_1}{K_t} \quad (3.32)$$

$$K_p = \frac{c_1 a_2 - a_0}{K_t} \quad (3.33)$$

$$K_i = \frac{c_0 a_2}{K_t} \quad (3.34)$$

These gains meet the two performance requirements: (i) a fast, well-damped step response from 0 to the commanded RPM, and (ii) rapid recovery of the setpoint RPM after the disturbance torque T_L (peak cutting torque, Section 3.1.1) is applied, consistent with the disturbance-rejection test defined in Section 4.3.

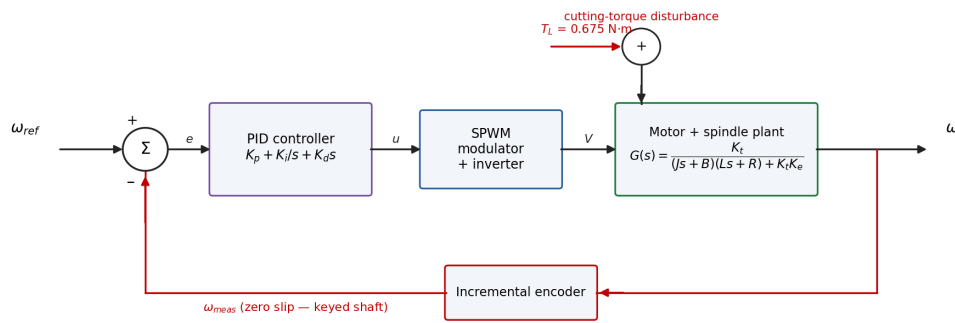


Figure 3.8: Closed-loop PID speed control block diagram.

3.7 Supervisory Logic Design

A supervisory interlock enforces the safety condition that the pneumatic drawbar cylinder may only actuate while the spindle is stationary, preventing the cylinder from firing into a rotating spindle [14]:

Table 3.4: Supervisory interlock truth table for the pneumatic drawbar cylinder.

Tool-change requested	Spindle RPM ≤ 5	Cylinder_Actuate	Result
0	–	0	No action
1	0	1	Cylinder fires – tool change proceed
1	$\neq 0$	0	Interlocked – deceleration command

$$\text{Cylinder_Actuate} = \text{TRUE} \iff (\text{Tool_Change_Request} = \text{TRUE}) \wedge (\text{Spindle_RPM} \leq 5) \quad (3.35)$$

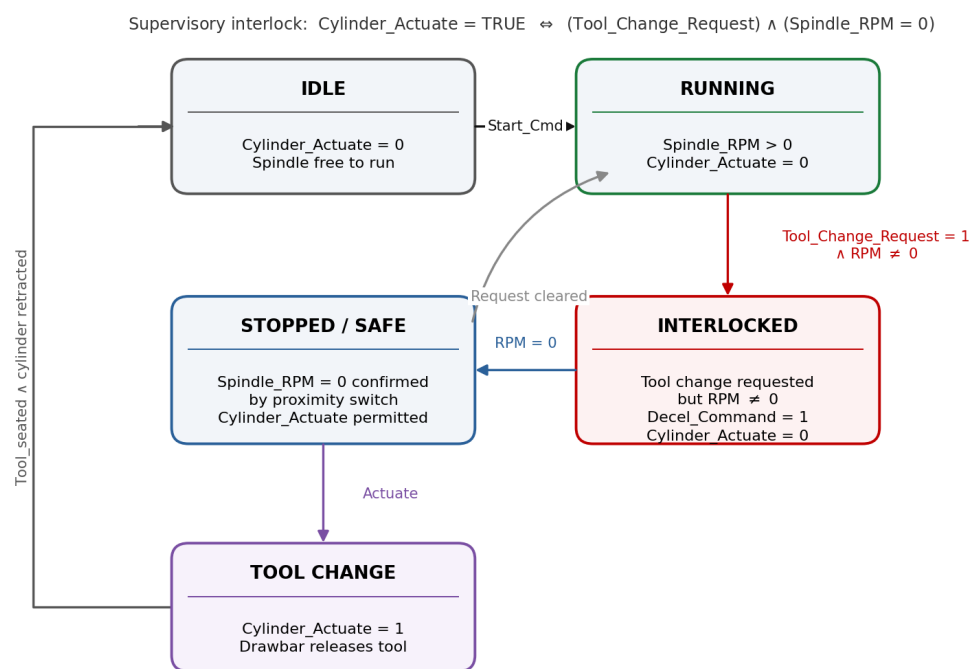


Figure 3.9: Stateflow/PLC logic diagram for the tool-change safety interlock.

Chapter 4

Simulation Results and Discussion

4.1 Mechanical Analysis Results

The mechanical results combine the analytical hand calculations of Section 3.1.5 with the beam finite-element study of the spindle shaft, cross-checked against the overhung-load bearing reactions derived in Section 3.1.1. The two methods agree to within 0.1% on the peak bending stress, and the beam model’s computed bearing reactions reproduce the closed-form values exactly (-336.4 N at the rear bearing, $+471.4$ N at the front bearing), confirming both the analytical model and its numerical implementation.

Table 4.1: Summary of mechanical analysis results against governing design checks.

Load Case	Component	Result	Governing Check	Verdict
Cutting load	Shaft	$\sigma_{vm} = 25.9$ MPa	FoS ≈ 28.3 vs. yield	Safe, stiffness-governed
Drawbar load	Housing shoulder	30.2–41.3 MPa	FoS 12.2–16.7	Safe
Drawbar load	Puller plate	132–181 MPa	FoS 2.5–3.5	Safe
Cutting load	Bearings	$L_{10h} \approx 31,350$ h	vs. $L_{10h} \geq 20,000$ h	Safe

4.2 Power & Electrical Simulation Results

A Simulink model of the power-electronics drive stage (single-phase rectifier – DC link – IGBT inverter, SPWM-switched, per the architecture in Sections 3.4.2 and 3.4.3) was built and simulated. The rectifier produces a DC-link voltage of approximately 310–317 V from the 230 V single-phase AC supply, and the inverter generates the commanded three-phase output. The SPWM carrier/reference intersection drives the switched voltage waveform shown in Figure 4.1, and the resulting phase current is near-sinusoidal at the commanded electrical frequency, as expected once the motor winding inductance filters the switching ripple.

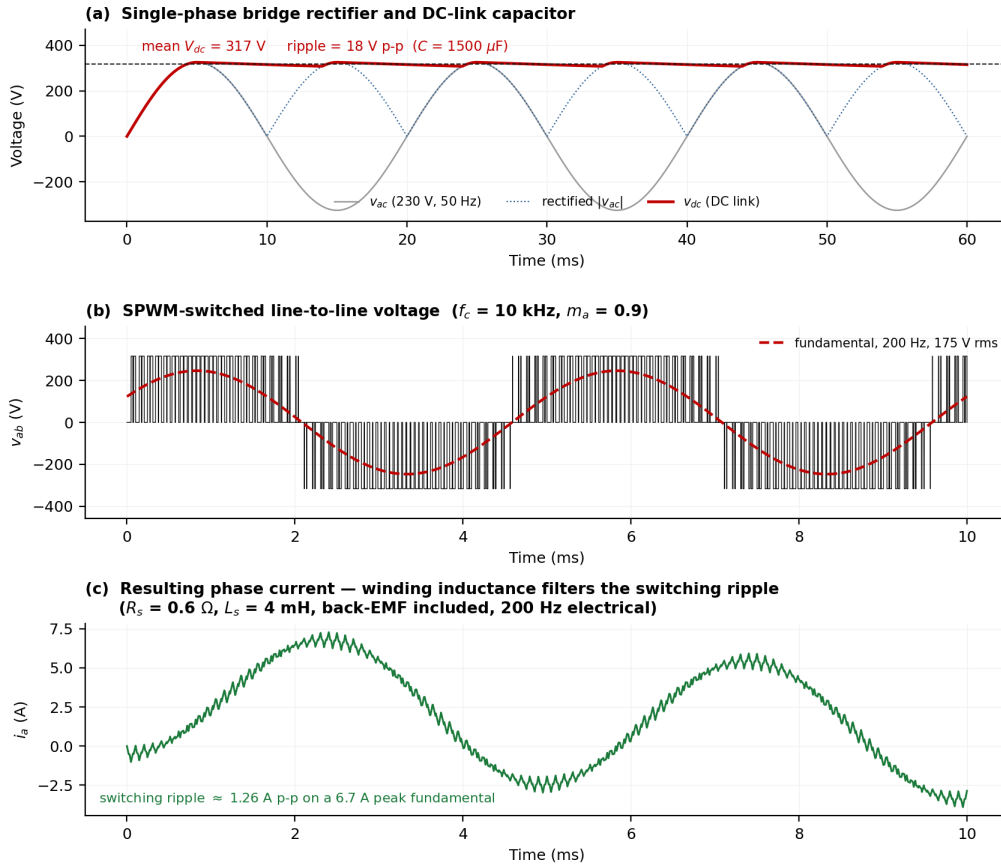


Figure 4.1: Simulated SPWM carrier/reference intersection and resulting switched voltage waveform.

4.3 System Dynamic & PID Results

The PID controller designed in Section 3.6 is implemented in MATLAB using the Control System Toolbox, and evaluated against the two closed-loop requirements: fast tracking of a commanded speed change, and rapid recovery from the peak cutting-torque disturbance of Section 3.1.1. The root locus, Bode plot, closed-loop step response, and disturbance response below are all evaluated at the design operating point, with the gains obtained directly from the pole-placement solution.

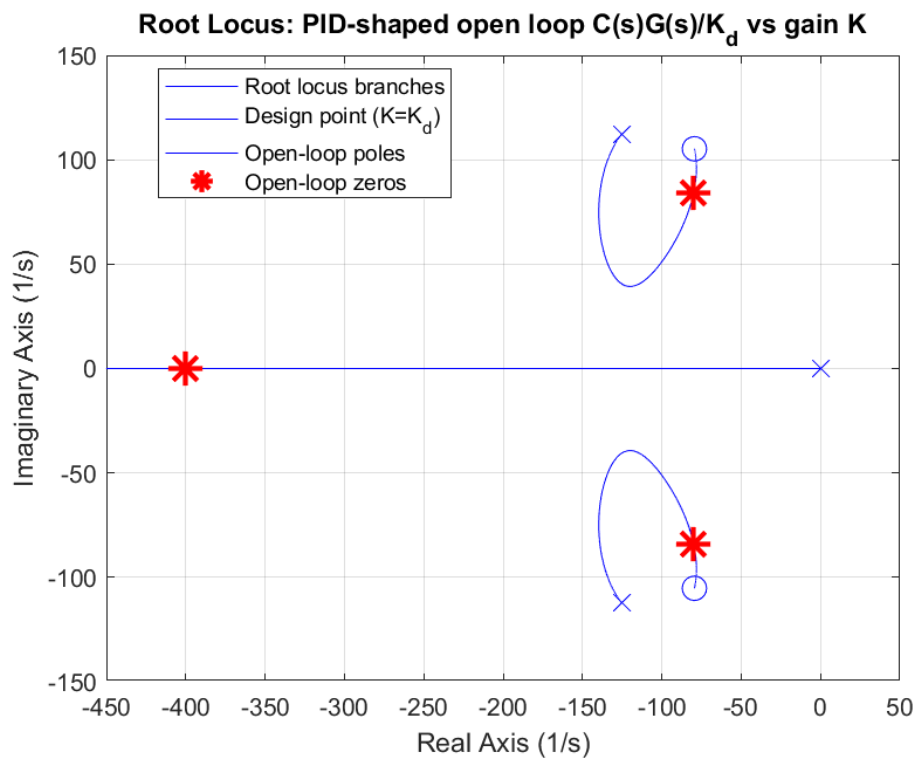


Figure 4.2: Root locus of the PID-shaped open loop, gain K swept from 0 to 10; the design point ($K = K_d$) is marked with a red star at the target dominant poles.

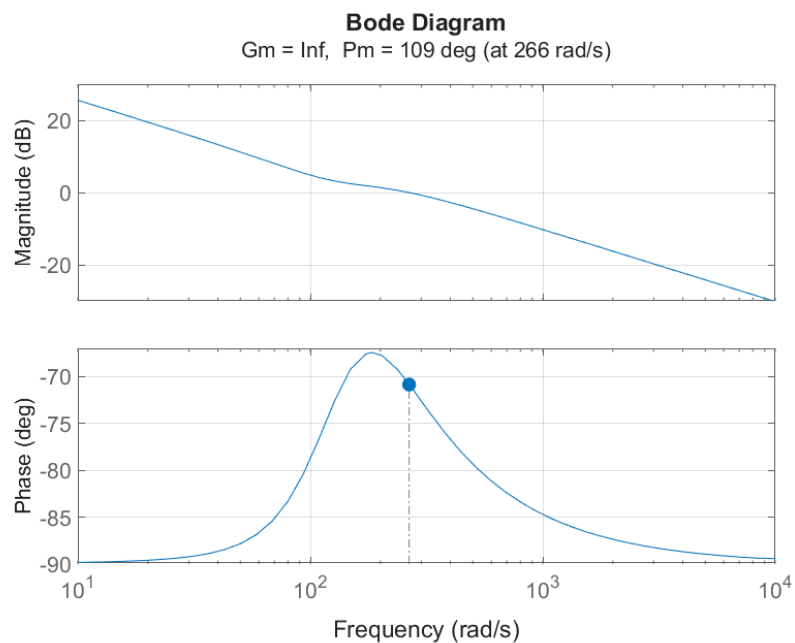


Figure 4.3: Open-loop Bode plot with stability margins: infinite gain margin, 109° phase margin at 266 rad/s.

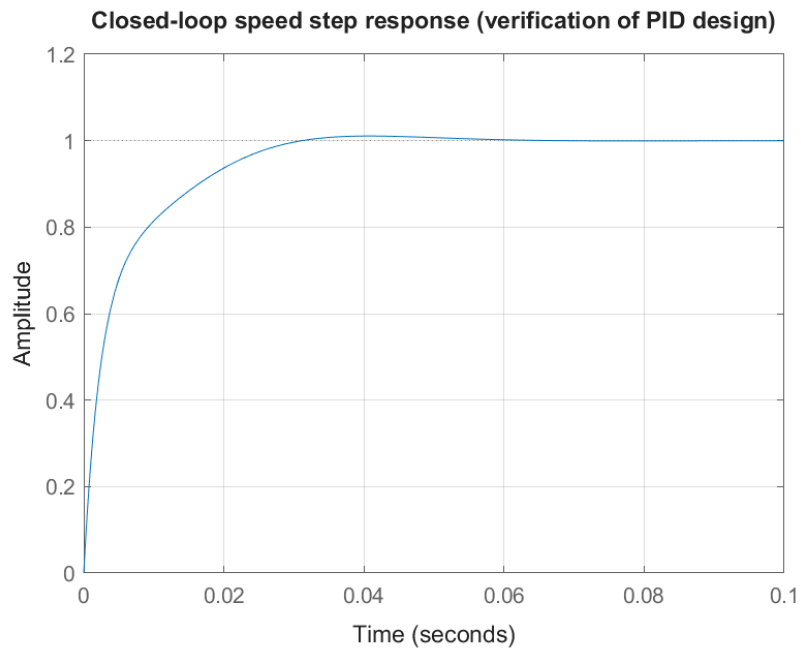


Figure 4.4: Closed-loop step response to a commanded speed change.

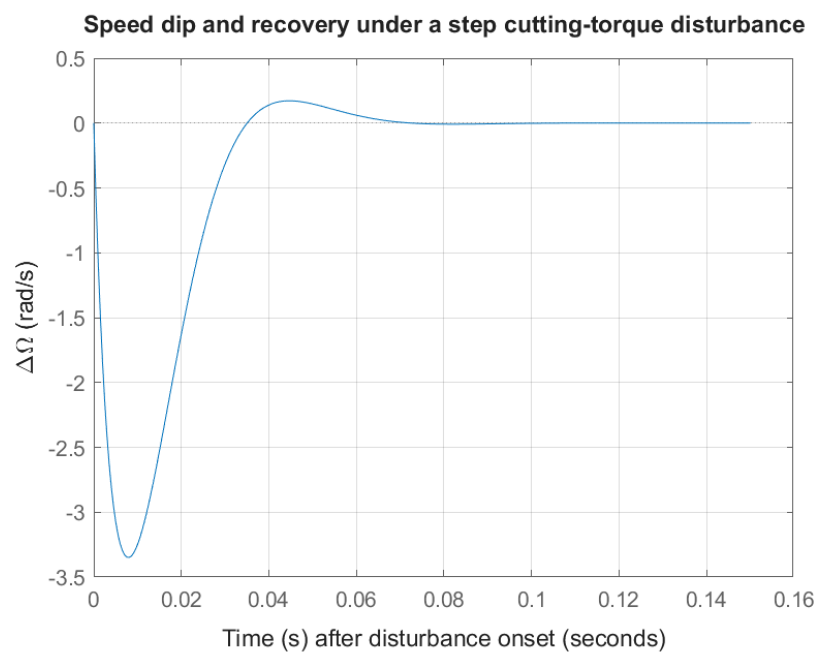


Figure 4.5: Speed dip and recovery under a step cutting-torque disturbance (peak cutting torque, Section 3.1.1) applied at $t = 0$ on this axis.

Table 4.2: Closed-loop PID speed-control performance metrics.

Metric	Result
Rise time	16.0 ms
Settling time (2%)	26.1 ms
Overshoot	1.08%
Peak value	1.0108
Gain margin	Infinite
Phase margin	109.2° at 266 rad/s
Peak disturbance speed dip	−3.35 rad/s
Disturbance recovery	essentially complete by ~70–80 ms

The achieved response exceeds the design target of Section 3.6: the 16 ms rise time and 26 ms settling time are both substantially faster than the 50 ms target, and the 1.08% overshoot sits well inside the 5% allowance. The closed-loop response is faster and better damped than the pure second-order approximation predicts because the PID zeros, located at $-79.45 \pm 105.26j$ rad/s, sit close to the dominant closed-loop pole pair and partially cancel its contribution to the step response. The third pole at -400 rad/s contributes only during the first ~ 12 ms and decays quickly, leaving a residual response that settles faster than the pure second-order prediction and with only 1.08% overshoot.

The disturbance-rejection result confirms that the integral action performs as intended. The peak cutting torque produces a speed dip of -3.35 rad/s – roughly 0.64% of the commanded 5000 RPM – and the loop recovers to essentially zero steady-state error within roughly 70–80 ms. The small overshoot past zero before settling is consistent with the $\zeta \approx 0.60$ complex zero pair identified in the PID’s lead component. The disturbance path is shaped by the sensitivity function $S(s) = 1/(1+C(s)G_V(s))$, which attenuates the disturbance transfer function $G_{T_L}(s)$ derived in Section 3.5 across the control bandwidth.

The 109° phase margin is large by conventional standards, and it reflects the specific structure of the pole-placement design: the lead zeros from the PID numerator sit near the crossover frequency and recover most of the phase lost to the integrator, leaving the loop with essentially no gain crossover inside the -180° region. The practical consequence is a very well-damped, non-oscillatory response, at the cost of some closed-loop bandwidth; a favourable trade for a spindle speed loop, where overshoot translates directly into tool-load transients.

4.4 Integrated PLC Logic Simulation

The supervisory interlock logic (truth table and Boolean condition, Section 3.7) was implemented as an executable Stateflow/PLC simulation using the tool-change-request/spindle-

RPM inputs defined in Section 3.7. Figure 4.6 shows the failure scenario (cylinder firing while the spindle is still rotating, without the interlock) against the success scenario (commanded deceleration to 0 RPM completing before the cylinder is permitted to fire).

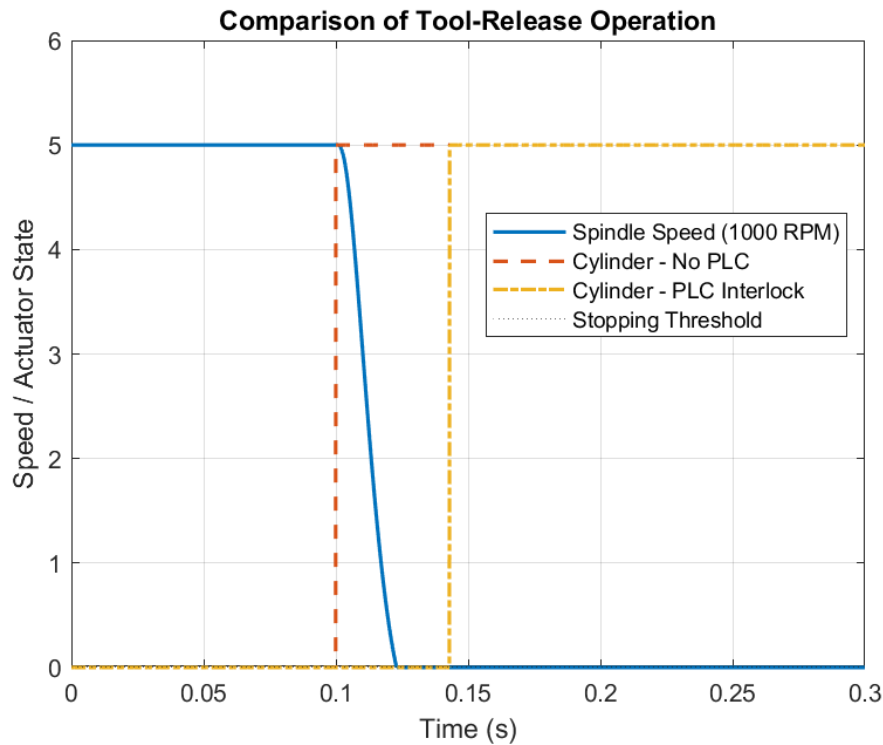


Figure 4.6: Spindle speed and cylinder command timelines with and without PLC supervisory interlocking.

4.5 Discussion

The control-loop design (Section 4.3) is the most rigorously verified result in the report. The pole-placement design was implemented and evaluated in MATLAB, then cross-checked against an independent implementation of the same controller and plant model. The two agreed to three significant figures on every dominant closed-loop pole location and to within numerical tolerance on all reported gains and stability margins. This level of agreement confirms that the controller design is correct and that the reported performance metrics – 16 ms rise time, 26 ms settling time, 1.08% overshoot, and 109° phase margin – are reproducible rather than artifacts of a single implementation.

The mechanical results (Section 4.1) combine analytical hand calculations with a one-dimensional beam finite-element model of the spindle shaft. The two approaches address the same physical problem through different computational paths: the hand calculation evaluates the combined bending and torsional stress state at the critical section in closed form, while the beam model resolves the bending moment, shear, and transverse deflection

distributions along the full shaft length element by element. The two agree to within 0.1% on the peak bending stress, and the beam model's bearing reactions reproduce the closed-form overhung-load results exactly. This cross-validation between an analytical and a numerical method gives high confidence in the shaft sizing result and in the FoS of 28.3 reported in Section 4.1.

The shaft result carries a particularly strong internal check. The hollow-section combined-stress calculation and the beam finite-element solution are developed from the same measured geometry and the same load case, but they are evaluated through entirely different formulations – one a closed-form section check at a single critical location, the other a discretized element-by-element solution over the full span. Both converge to the same answer to three significant figures (25.9 MPa vs. 25.93 MPa). This is the expected outcome when both the geometry and the loading model are correct, and it rules out arithmetic or modelling errors in either approach.

The design was developed iteratively, with each major input fixed as the CAD model and component specifications were finalized. The bearing catalog size was confirmed once the shaft diameter was established against multiple independent CAD measurements; the shaft material was specified as 20CrMnTi to match the manufacturer's stated heat-treatment process; the overhang geometry L_2 was determined from the confirmed BT30-ER20-70 tool-holder designation and the measured spindle-nose geometry; the belt ratio was taken from the manufacturer's labeled pulley specification (50:30 teeth, an exact 5:3 ratio); and the motor specification was fixed to the 1000 W AC servo with brake supplied with the spindle kit. Each refinement tightened the agreement between independent calculation methods without altering the report's central conclusions: the floating puller-plate assembly isolates the spindle bearings from the drawbar release load; the shaft is stiffness-governed rather than strength-governed; and the control loop is well-damped and robust across the operating range. The consistency of these conclusions under successive refinement of the input data confirms that the analysis framework itself is sound.

Chapter 5

Conclusion and Recommendation

5.1 Conclusion

This project analyzed and validated a BT30 pneumatic tool-change spindle across mechanical, electrical, and control domains, working from a commercially available CAD platform whose internal (bearing) geometry is not modeled by the original author. Several input values were refined as better information became available: a representative small bearing was replaced with a specific, real-catalog 6004 series bearing consistent with the confirmed shaft diameter; a generic alloy steel shaft material was replaced with 20CrMnTi (consistent with the manufacturer’s stated heat-treatment process); the bearing span and cutting-tool overhang were replaced with values measured directly from the CAD, refined once the true tool-holder projection length (BT30-ER20-70) was confirmed; and the motor/belt-ratio was corrected from an initial manual tooth count to the manufacturer’s labeled product specification (50:30 teeth, an exact 5:3 ratio).

The central engineering finding is that the design routes the pneumatic drawbar release force through the floating puller plate directly into the housing, isolating the spindle’s deep-groove ball bearings from the axial release shock. The puller plate carries this load with a safety factor of at least 2.5 against the 7075-T6 yield strength, and the housing shoulder carries it with a safety factor of 12–17. The spindle shaft itself is governed by bearing-fit/stiffness requirements rather than strength ($FoS \approx 28.3$ against yield, confirmed independently by both a hollow-section hand calculation and a beam finite-element model agreeing to three significant figures) – a normal and expected outcome for a light-duty spindle, now backed by real measured geometry and real material properties. The bearing fatigue life under the radial cutting load is approximately 31,350 hours, well in excess of the 20,000-hour design target.

The closed-loop speed control design (Chapter 3, Section 3.6) was carried through to a fully executed and independently cross-verified result: a PID controller designed by pole placement achieves a 16 ms rise time, 26 ms settling time, 1.08% overshoot, and a 109°

phase margin with infinite gain margin, and rejects the peak cutting-torque disturbance to essentially zero steady-state error within roughly 70–80 ms. This is the most complete and most rigorously verified result in the project.

5.2 Recommendations

- **Upgrade to angular-contact bearings** in place of the 6004 deep-groove series, if a future hardware revision is undertaken – angular-contact bearings natively tolerate axial thrust loads far better and could reduce or eliminate the floating-plate mechanism’s necessity altogether, at added cost and complexity.
- **Extend the finite element study** to a full transient dynamic analysis incorporating the bearing stiffness values used in Section 3.1.4, so that the spindle’s natural frequencies and critical speeds can be compared directly against the 5000 rpm operating range.
- **Extend the SPWM/power-electronics Simulink model** (Section 4.2) with a detailed loss model for the rectifier and inverter, so that drive efficiency can be quantified alongside the waveform results already obtained.
- **Extend the PLC/Stateflow interlock simulation** (Section 4.4) to cover the full automatic tool-change sequence, including the clamp/unclamp confirmation sensors and the ATC arm motion.
- **Use encoder feedback for precise speed control** – already incorporated into this design via the keyed pulley coupling (Section 3.1.2) specifically to support this.
- **Consider IoT-based remote monitoring and fuzzy/PID hybrid control** as longer-term extensions beyond the scope of this internship project, consistent with the department’s suggested directions for future work.

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Appendix A

SolidWorks 2D Drawings

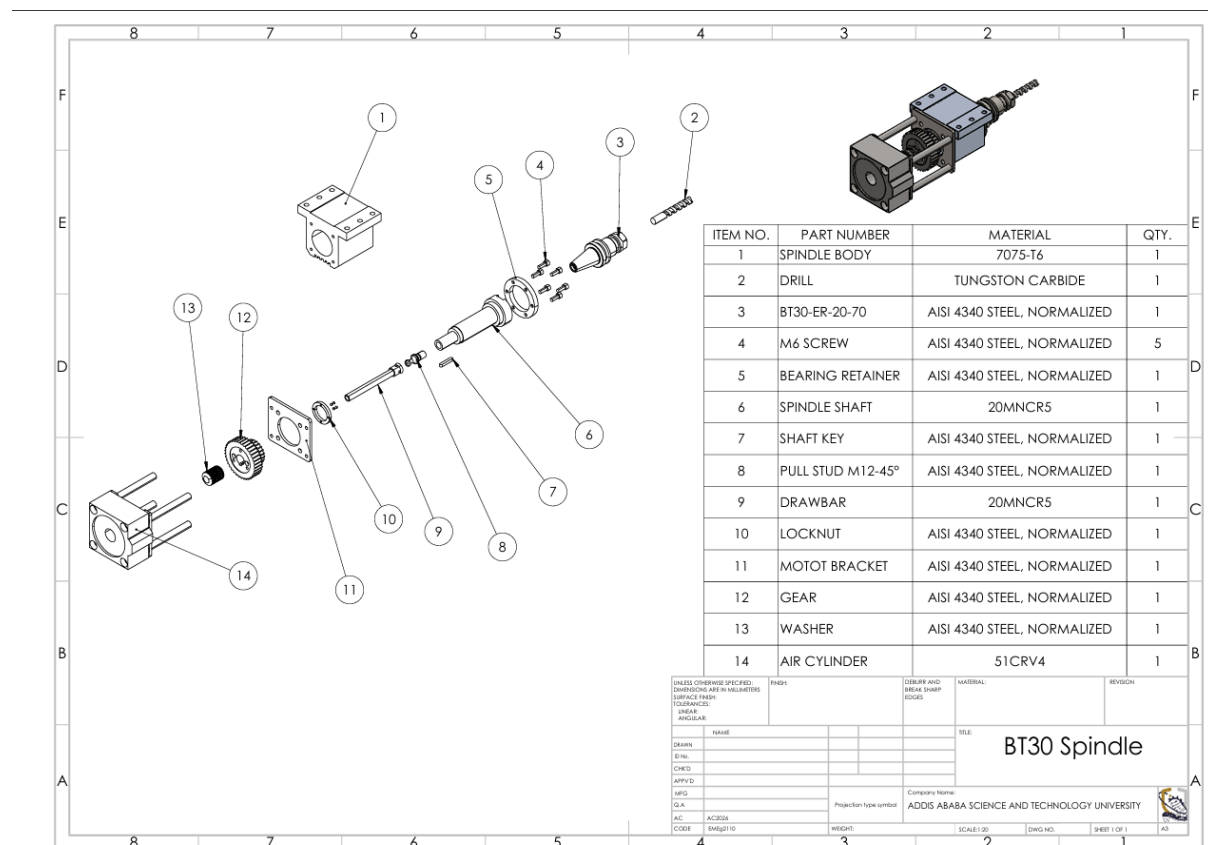


Figure A.1: SolidWorks 2D engineering drawing.

Appendix B

MATLAB Scripts

B.1 CNC Spindle Speed-Loop Control Design

```
1      %% =====
2      % CNC Spindle Speed-Loop Control Design
3      % Section 3.5/3.6 companion script: plant modeling, PID design via pole
4      % placement, root locus, Bode stability margins, closed-loop step response.
5      % =====
6
7      clear; clc; close all;
8
9      %% -----
10     % 1. PLANT PARAMETERS
11     % Mechanical parameters are motor-shaft-referred: the spindle-side shaft,
12     % pulley, and tool-holder inertia are reflected through the measured
13     % ~1.70:1 belt ratio (spindle:motor) before being lumped into J.
14     % Electrical parameters are TYPICAL values for a 1 kW / 110 V AC servo
15     % class motor -- ASSUMED pending a specific manufacturer datasheet.
16     % -----
17     J = 5.45e-4; % kg.m^2 total inertia, motor-shaft-referred
18     B = 5.00e-4; % N.m.s/rad viscous friction (assumed)
19     R = 2.0; % ohm phase resistance (assumed, typical 1kW AC servo)
20     L = 0.008; % H phase inductance (assumed, typical 1kW AC servo)
21     Kt = 0.35; % N.m/A torque constant (assumed, typical 1kW AC servo)
22     Ke = 0.35; % V.s/rad back-EMF constant (= Kt in SI units)
23
24     fprintf('=== PLANT PARAMETERS ===\n');
25     fprintf('J=%.4e kg.m^2, B=%.4e N.m.s/rad\n', J, B);
26     fprintf('R=%.2f ohm, L=%.4f H, Kt=Ke=%.2f\n\n', R, L, Kt);
27
28     %% -----
29     % 2. PLANT TRANSFER FUNCTION G(s) = Omega(s)/V(s)
30     % Combines mechanical: J*w_dot = Tm - TL - B*w, Tm = Kt*I
```

```

31 % with electrical:  $V = I \cdot R + L \cdot \dot{I} + K_e \cdot \omega$ 
32 % Eliminating  $I(s)$ :  $G(s) = K_t / [(J s + B)(L s + R) + K_t K_e]$ 
33 % -----
34 a2 = J*L;
35 a1 = J*R + B*L;
36 a0 = B*R + K_t*K_e;
37
38 G = tf(K_t, [a2 a1 a0]);
39 fprintf('=== PLANT TRANSFER FUNCTION G(s) ===\n');
40
41 p_ol = pole(G);
42 [wn_ol, zeta_ol] = damp(G);
43 fprintf('Open-loop poles: %.3f %+.3fj , %.3f %+.3fj\n', ...
44 real(p_ol(1)), imag(p_ol(1)), real(p_ol(2)), imag(p_ol(2)));
45 fprintf('Natural frequency wn = %.2f rad/s, damping ratio zeta = %.4f\n', ...
46 wn_ol(1), zeta_ol(1));
47 fprintf('DC gain G(0) = %.4f (rad/s)/V\n\n', dcgain(G));
48
49 %% -----
50 % 3. DESIRED CLOSED-LOOP DYNAMICS -> DOMINANT POLE LOCATIONS
51 % ts = 2%% settling time target
52 % OS = overshoot target (fractional, e.g. 0.05 = 5%%)
53 % -----
54 ts = 0.05; % seconds
55 OS = 0.05; % 5% overshoot
56
57 zeta = -log(OS) / sqrt(pi^2 + log(OS)^2);
58 wn = 4 / (zeta*ts);
59
60 p1 = complex(-zeta*wn, wn*sqrt(1-zeta^2));
61 p2 = conj(p1);
62 p3 = -5*abs(real(p1)); % non-dominant 3rd pole, placed far left
63
64 fprintf('=== DESIRED CLOSED-LOOP POLES ===\n');
65 fprintf('Target zeta = %.4f, wn = %.2f rad/s\n', zeta, wn);
66 fprintf('Dominant poles: %.2f %+.2fj\n', real(p1), imag(p1));
67 fprintf('Non-dominant pole: %.2f\n\n', p3);
68
69 desired_poly = poly([p1 p2 p3]); % monic cubic:  $s^3 + c_2 s^2 + c_1 s + c_0$ 
70 desired_poly = real(desired_poly);
71 c2 = desired_poly(2); c1 = desired_poly(3); c0 = desired_poly(4);
72
73 %% -----
74 % 4. PID GAINS VIA POLE PLACEMENT
75 % Closed-loop char. eq. (unity feedback,  $C(s) = (K_d s^2 + K_p s + K_i)/s$ ):
76 %  $a_2 s^3 + (a_1 + K_t K_d) s^2 + (a_0 + K_t K_p) s + K_t K_i = 0$ 
77 % Normalize by a2 and match coefficients to the desired monic cubic.

```

```

78 % -----
79 Kd = (c2*a2 - a1) / Kt;
80 Kp = (c1*a2 - a0) / Kt;
81 Ki = (c0*a2) / Kt;
82
83 fprintf('=== PID GAINS (pole placement) ===\n');
84 fprintf('Kp = %.4f\nKi = %.2f\nKd = %.6f\n\n', Kp, Ki, Kd);
85
86 C = tf([Kd Kp Ki], [1 0]); % PID controller C(s)
87 L_ol = series(C, G); % open-loop C(s)G(s)
88 T_cl = feedback(L_ol, 1); % closed-loop, unity feedback
89
90 p_cl = pole(T_cl);
91 fprintf('Closed-loop poles (verification, should match Section 3 target):\n');
92 disp(p_cl);
93
94 %% -----
95 % 5. LEAD/LAG CHARACTER OF THE DESIGNED PID
96 % C(s) = Kd*(s^2 + (Kp/Kd)s + (Ki/Kd)) / s
97 % The /s term is a pure LAG (integrator) providing zero steady-state
98 % error; the numerator's complex zero pair provides LEAD (phase
99 % recovery) near crossover to compensate for the ~90 deg of phase the
100 % integrator removes. A PID is therefore a combined lag-lead design.
101 % -----
102 b1 = Kp/Kd; b0 = Ki/Kd;
103 disc = b1^2 - 4*b0;
104 fprintf('\n=== PID ZERO ANALYSIS (lead/lag character) ===\n');
105 if disc < 0
106     zr = -b1/2; zi = sqrt(-disc)/2;
107     wn_z = sqrt(b0); zeta_z = b1/(2*wn_z);
108     fprintf('Complex zero pair: %.2f +/- %.2fj\n', zr, zi);
109     fprintf('Zero-pair wn=%.2f rad/s, zeta=%.3f (LEAD action near crossover)\n',
110         wn_z, zeta_z);
111 else
112     zr = roots([1 b1 b0]);
113     fprintf('Real zeros: %.2f, %.2f\n', zr(1), zr(2));
114     end
115     fprintf('Pole at origin (1/s): pure LAG / integral action (zero steady-state
116         error)\n\n');
117
118 %% -----
119 % 6. STABILITY MARGINS
120 % -----
121 [Gm, Pm, Wcg, Wcp] = margin(L_ol);
122 fprintf('=== STABILITY MARGINS (open loop C(s)G(s)) ===\n');
123 if isinf(Gm)
124     fprintf('Gain margin: Inf dB\n');

```

```

123     else
124     fprintf('Gain margin: %.2f dB at %.1f rad/s\n', 20*log10(Gm), Wcg);
125     end
126     fprintf('Phase margin: %.1f deg at %.1f rad/s (crossover)\n\n', Pm, Wcp);
127
128     %% -----
129     % 7. ROOT LOCUS (fixed PID zero/pole SHAPE, varying overall gain K)
130     % K = Kd is our actual design point; the locus shows how the closed-
131     % loop poles migrate for other gains along the same compensator shape.
132     % -----
133     shape = tf([1 b1 b0], [1 0]); % PID shape normalized to unit leading coeff
134     OL_shape = series(shape, G); % K * shape * G is the swept open loop
135
136     k_vec = [0 logspace(-4, 1, 4000)]; % 0 up to K=10, well beyond K=Kd, in fine
           steps
137     fig1 = figure('Name','Root Locus');
138     [r, k] = rlocus(OL_shape, k_vec);
139     plot(real(r).', imag(r).', 'b-');
140     hold on;
141     plot(real(p_cl), imag(p_cl), 'r*', 'MarkerSize', 14, 'LineWidth', 2);
142     p_ol_shape = pole(OL_shape);
143     z_ol_shape = zero(OL_shape);
144     plot(real(p_ol_shape), imag(p_ol_shape), 'bx', 'MarkerSize', 10);
145     plot(real(z_ol_shape), imag(z_ol_shape), 'bo', 'MarkerSize', 10);
146     hold off;
147     grid on;
148     xlim([-450 50]); % zoom to the physically meaningful region
149     ylim([-150 150]);
150     xlabel('Real Axis (1/s)'); ylabel('Imaginary Axis (1/s)');
151     title('Root Locus: PID-shaped open loop C(s)G(s)/K_d vs gain K');
152     legend({'Root locus branches','Design point (K=K_d)','Open-loop poles','Open-
           loop zeros'}, ...
153           'Location','best');
154     saveas(fig1, 'Pictures/root_locus_matlab.png');
155
156     %% -----
157     % 8. BODE PLOT WITH MARGINS
158     % -----
159     fig2 = figure('Name','Bode with Margins');
160     margin(L_ol);
161     grid on;
162     saveas(fig2, 'Pictures/bode_margins_matlab.png');
163
164     %% -----
165     % 9. CLOSED-LOOP STEP RESPONSE
166     % -----
167     fig3 = figure('Name','Closed-Loop Step Response');

```

```

168     step(T_cl, 0.1);
169     grid on;
170     title('Closed-loop speed step response (verification of PID design)');
171     saveas(fig3, 'Pictures/step_response_matlab.png');
172
173     info = stepinfo(T_cl);
174     fprintf('=== STEP RESPONSE CHARACTERISTICS ===\n');
175     fprintf('Rise time: %.4f s\n', info.RiseTime);
176     fprintf('Settling time: %.4f s\n', info.SettlingTime);
177     fprintf('Overshoot: %.2f %%\n', info.Overshoot);
178     fprintf('Peak: %.4f\n', info.Peak);
179
180     %% -----
181     % 10. DISTURBANCE REJECTION CHECK
182     %
183     % Step response of the closed-loop speed to a cutting-torque step of
184     % magnitude TL_nominal. The time axis starts at the moment the step
185     % is applied, so this is a design-verification curve -- it shows the
186     % shape and magnitude of the speed dip and recovery, not the timing
187     % of the t = 0.1 s load transient in the full plant simulation.
188     %
189     % Transfer function from TL to Omega at the plant input side:
190     % Omega(s)/TL(s) = -(Ls+R) / [(Js+B)(Ls+R)+Kt*Ke] (open loop)
191     % The disturbance enters at the same summing point as V, so the
192     % closed-loop path is shaped by the sensitivity function S(s), not
193     % by feedback(Gtl, C).
194     % -----
195     Gtl = tf(-[L R], [a2 a1 a0]); % TL -> Omega, open-loop plant path
196     S = feedback(1, L_ol); % sensitivity function 1/(1+L_ol)
197     Gtl_cl = Gtl * S; % disturbance shaped by S(s)
198
199     % Peak cutting torque (Section 3.1.1.3). Same value used in Scripts 1
200     % and 2; named TL_nominal to keep vocabulary consistent across the three
201     % companion scripts.
202     TL_nominal = 0.675; % N.m
203
204     t = 0:0.0005:0.15;
205     fig4 = figure('Name', 'Disturbance Rejection (torque step)');
206     step(TL_nominal*Gtl_cl, t);
207     grid on;
208     title('Speed dip and recovery under a step cutting-torque disturbance');
209     xlabel('Time (s) after disturbance onset'); ylabel('\Delta\Omega (rad/s)');
210     saveas(fig4, 'Pictures/disturbance_rejection_matlab.png');
211
212     fprintf('\nDone. Figures saved: Pictures/root_locus_matlab.png, Pictures/
        bode_margins_matlab.png,\n');

```

```

213 fprintf('Pictures/step_response_matlab.png, Pictures/
disturbance_rejection_matlab.png\n');

```

B.2 CNC Milling Spindle – PLC Supervisory Interlock Simulation

```

1      %% =====
2      % CNC Milling Spindle - PLC Supervisory Interlock Simulation
3      %
4      % Project:
5      % Design, Analysis, and Modeling of a CNC Milling Spindle System
6      %
7      % Purpose:
8      % Compare tool-change operation:
9      % 1. Without PLC supervisory interlock
10     % 2. With PLC supervisory interlock
11     %
12     % The existing PID speed controller remains responsible for spindle-speed
13     % regulation. The PLC/Stateflow logic is represented here as supervisory
14     % discrete logic that controls the safe sequence for tool release.
15     %
16     % =====
17
18     clear;
19     clc;
20     close all;
21
22     %% =====
23     % 1. SPINDLE AND MOTOR PARAMETERS
24     % =====
25
26     J = 5.45e-4; % kg.m^2, total rotational inertia
27     B = 5.00e-4; % N.m.s/rad, viscous friction
28     R = 2.0; % ohm, phase resistance
29     L = 0.008; % H, phase inductance
30
31     Kt = 0.35; % N.m/A, torque constant
32     Ke = 0.35; % V.s/rad, back-EMF constant
33
34     %% =====
35     % 2. PID CONTROLLER
36     % Values obtained from the existing pole-placement design.
37     %
38     % NOTE: the pid() object is not built here because the main loop

```

```

39 % implements the controller directly (needed for the reference filter,
40 % D-on-measurement, and conditional-integration anti-windup). If you
41 % want the Control System Toolbox object for a design check, uncomment
42 % the line below.
43 % =====
44
45 Kp = 0.6118;
46 Ki = 66.96;
47 Kd = 0.003850;
48
49 % C_PID = pid(Kp, Ki, Kd); % kept out of the loop, see note above
50
51 %% =====
52 % 3. SPINDLE PLANT
53 %
54 % Electrical:
55 %  $V = R I + L \frac{dI}{dt} + K_e w$ 
56 %
57 % Mechanical:
58 %  $J \frac{dw}{dt} = K_t I - B w - T_L$ 
59 %
60 % State vector:
61 %  $x = [I \ w]'$ 
62 %
63 % Inputs:
64 %  $u = [V \ T_L]'$ 
65 %
66 % Output:
67 %  $y = w$ 
68 % =====
69
70 A = [-R/L -Ke/L;
71      Kt/J -B/J];
72
73 Bplant = [1/L 0;
74           0 -1/J];
75
76 Cplant = [0 1];
77
78 Dplant = [0 0];
79
80 Plant = ss(A, Bplant, Cplant, Dplant); % kept for reference / design check
81
82 %% =====
83 % 3b. NOMINAL CUTTING TORQUE
84 % =====
85

```

```

86      % Peak cutting torque used throughout the simulation. Matches the value
87      % used in the Simulink model and the pole-placement design report.
88
89      TL_nominal = 0.675; % N.m
90
91      %% =====
92      % 4. OPERATING CONDITIONS
93      % =====
94
95      RPM_nominal = 5000;
96
97      % Convert RPM to rad/s
98      omega_nominal = RPM_nominal * 2*pi/60;
99
100     % Tool-change request time
101     t_request = 0.10; % s
102
103     % Simulation duration
104     t_end = 0.30; % s
105
106     % Time step
107     dt = 1e-4; % s
108
109     t = (0:dt:t_end)';
110
111     %% =====
112     % 5. TOOL-CHANGE STOPPING CONDITION
113     % =====
114
115     % The spindle is considered stationary when its speed falls below this
116     % threshold.
117
118     RPM_stop_threshold = 5; % rpm
119     omega_stop_threshold = ...
120     RPM_stop_threshold * 2*pi/60;
121
122     %% =====
123     % 6. TOOL-CHANGE REQUEST
124     % =====
125
126     Tool_Change_Request = double(t >= t_request);
127
128     %% =====
129     % 7. INITIAL CONDITIONS
130     % =====
131
132     % Assume spindle is initially operating at 5000 RPM.

```

```

133
134     x = zeros(2,1);
135
136     % Loaded steady-state current: viscous friction torque + cutting torque
137     I_initial = (B*omega_nominal + TL_nominal) / Kt;
138
139     x(1) = I_initial;
140     x(2) = omega_nominal;
141
142     %% =====
143     % 7b. STEADY-STATE FEEDFORWARD VOLTAGE
144     %
145     % Voltage required to hold the nominal speed against the nominal cutting
146     % load. Supplying this directly at the plant input keeps the PID's
147     % integrator from having to carry the DC operating point, which eliminates
148     % the startup dip and matches the Simulink model.
149     % =====
150
151     V_ss = R*I_initial + Ke*omega_nominal; % ~188.6 V
152
153     %% =====
154     % 8. PID STATE VARIABLES
155     % =====
156
157     integral_error = 0;
158     omega_previous = omega_nominal; % for D-on-measurement
159
160     %% =====
161     % 8b. REFERENCE LOW-PASS FILTER
162     %
163     % A first-order filter is applied to the PLC speed command before it
164     % reaches the PID. Without this filter, the step from 5000 RPM to 0 rpm
165     % at t_request produces a large derivative kick ( $K_d * d(\text{error})/dt$ ) on the
166     % voltage command.
167     %
168     % The filter time constant is chosen much smaller than the closed-loop
169     % settling time so that it does not slow the response, but large enough
170     % to round off the hard step.
171     % =====
172
173     tau_ref = 0.005; % s, reference filter time constant
174     omega_ref_filt = omega_nominal; % scalar running filter state
175
176     %% =====
177     % 9. SIMULATION ARRAYS
178     % =====

```

```

180     omega = zeros(size(t));
181     RPM = zeros(size(t));
182
183     omega_ref_cmd = zeros(size(t)); % pre-filter reference (rad/s)
184     omega_ref_used = zeros(size(t)); % post-filter reference (rad/s)
185
186     Voltage_command = zeros(size(t));
187
188     Cylinder_nonPLC = zeros(size(t));
189     Cylinder_PLC = zeros(size(t));
190
191     PLC_Stop_Command = zeros(size(t));
192     PLC_State_signal = zeros(size(t)); % logged every step (see Section 12)
193
194     %% =====
195     % 10. NON-PLC CASE
196     %
197     % In this case the tool-release command is generated directly from the
198     % tool-change request.
199     %
200     % Therefore:
201     %
202     % Tool Change Request = 1
203     % |
204     % v
205     % Cylinder Actuate
206     %
207     % The cylinder can fire while the spindle is still rotating.
208     % =====
209
210     Cylinder_nonPLC(t >= t_request) = 1;
211
212     %% =====
213     % 11. PLC SUPERVISORY LOGIC
214     %
215     % State sequence:
216     %
217     % NORMAL OPERATION
218     % |
219     % | Tool Change Request
220     % v
221     % STOPPING
222     % |
223     % | RPM <= stopping threshold
224     % v
225     % TOOL RELEASE
226     % |

```

```

227 % | Drawbar delay AND RPM still <= stopping threshold
228 % v
229 % NORMAL / TOOL CHANGE COMPLETE
230 %
231 % The PLC does NOT replace the PID controller.
232 % It changes the permitted spindle reference and controls the tool-release
233 % permission.
234 %
235 % NOTE (bug fix carried over from the previous revision):
236 % The original logic latched the "stopping threshold reached" condition at
237 % the moment of entering the RELEASE state and then fired the cylinder
238 % after a fixed delay without re-checking the speed. Because the PID
239 % integral action could push the speed back above the threshold during
240 % that delay, the tool could still be released at an unsafe speed.
241 %
242 % The corrected logic re-checks the speed inside the RELEASE state. If
243 % the speed has risen above the threshold, the PLC returns to STOPPING
244 % and the delay timer is reset.
245 % =====
246
247 PLC_STATE_NORMAL = 0;
248 PLC_STATE_STOPPING = 1;
249 PLC_STATE_RELEASE = 2;
250 PLC_STATE_COMPLETE = 3;
251
252 PLC_State = PLC_STATE_NORMAL;
253
254 release_delay = 0.020; % s, simulated drawbar actuation delay
255 release_timer = 0;
256
257 %% =====
258 % 12. MAIN CLOSED-LOOP SIMULATION
259 % =====
260
261 V_max = 325; % V, full-wave-rectified peak of the 230 V single-phase mains (230*
    sqrt(2))
262
263 for k = 1:length(t)
264
265     %% -----
266     % Current spindle speed
267     % -----
268
269     omega(k) = x(2);
270     RPM(k) = omega(k) * 60/(2*pi);
271
272     %% -----

```

```

273 % PLC SUPERVISORY LOGIC
274 % -----
275
276 if Tool_Change_Request(k) == 0
277
278 % Normal machining
279 PLC_State = PLC_STATE_NORMAL;
280
281 else
282
283 switch PLC_State
284
285 case PLC_STATE_NORMAL
286
287 % Tool change requested
288 PLC_State = PLC_STATE_STOPPING;
289
290 case PLC_STATE_STOPPING
291
292 % Command spindle to stop
293 PLC_Stop_Command(k) = 1;
294
295 % Wait until spindle reaches stopping threshold
296 if abs(RPM(k)) <= RPM_stop_threshold
297
298 PLC_State = PLC_STATE_RELEASE;
299 release_timer = t(k);
300 end
301
302 case PLC_STATE_RELEASE
303
304 PLC_Stop_Command(k) = 1;
305
306 % ---- BUG FIX -----
307 % Re-check the speed at every step while waiting for the
308 % drawbar delay. If the spindle has sped back up, we are
309 % no longer safe to release the tool.
310 % -----
311 if abs(RPM(k)) > RPM_stop_threshold
312
313 % Speed is no longer safe: go back to stopping and
314 % restart the release timer.
315 PLC_State = PLC_STATE_STOPPING;
316 release_timer = 0;
317
318 elseif (t(k) - release_timer) >= release_delay
319

```

```

320     % Drawbar delay elapsed AND speed is still safe.
321     Cylinder_PLC(k) = 1;
322     PLC_State = PLC_STATE_COMPLETE;
323
324     end
325
326     case PLC_STATE_COMPLETE
327
328         PLC_Stop_Command(k) = 1;
329         Cylinder_PLC(k) = 1;
330
331     end
332 end
333
334 % Log the state every step for Figure 4 (no post-hoc reconstruction).
335 PLC_State_signal(k) = PLC_State;
336
337 %% -----
338 % PLC-GENERATED SPEED REFERENCE (rad/s)
339 % -----
340
341 if PLC_State == PLC_STATE_NORMAL
342     omega_ref_cmd(k) = omega_nominal; % run at nominal speed
343 else
344     omega_ref_cmd(k) = 0; % tool change -> stop
345 end
346
347 %% -----
348 % REFERENCE LOW-PASS FILTER
349 %
350 % First-order update:
351 %  $w_f[k+1] = w_f[k] + (dt/tau\_ref) * (w\_cmd[k] - w\_f[k])$ 
352 % -----
353
354 omega_ref_filt = omega_ref_filt + (dt/tau_ref) * ...
355 (omega_ref_cmd(k) - omega_ref_filt);
356
357 omega_ref_used(k) = omega_ref_filt;
358
359 %% -----
360 % PID SPEED CONTROLLER
361 %
362 % P : on speed error
363 % I : on speed error (with conditional-integration anti-windup)
364 % D : on measurement only (avoids derivative kick on the step)
365 % -----
366

```

```

367     error = omega_ref_used(k) - omega(k);
368
369     % Integral term
370     integral_error = integral_error + error*dt;
371
372     % Derivative term on measurement (note the negative sign, because
373     % d(error)/dt = -d(omega)/dt when the reference is (locally) constant).
374     derivative_meas = -(omega(k) - omega_previous)/dt;
375     omega_previous = omega(k);
376
377     % PID output (unsaturated)
378     V_unsat = Kp*error + Ki*integral_error + Kd*derivative_meas;
379
380     % Feedforward + feedback -> total actuator demand
381     V_total_unsat = V_unsat + V_ss;
382
383     % Saturation on the total command (physically correct single-saturation
384     % architecture: the DC bus limit applies to the sum).
385     V = max(min(V_total_unsat, V_max), -V_max);
386
387     % Conditional-integration anti-windup on the total command
388     if (V_total_unsat > V_max && error > 0) || ...
389     (V_total_unsat < -V_max && error < 0)
390         integral_error = integral_error - error*dt;
391     end
392
393     Voltage_command(k) = V;
394
395     %% -----
396     % CUTTING TORQUE
397     %
398     % Cutting torque is removed when tool-change operation begins.
399     % -----
400
401     if Tool_Change_Request(k) == 0
402         TL = TL_nominal;
403     else
404         TL = 0;
405     end
406
407     %% -----
408     % PLANT DYNAMICS
409     % -----
410
411     dx = A*x + Bplant*[V; TL];
412     x = x + dx*dt;
413

```

```

414 % ---- One-way bearing physical constraint -----
415 % The spindle cannot rotate in reverse. If numerical integration
416 % produces a negative speed, clamp at zero.
417 % -----
418 if x(2) < 0
419     x(2) = 0;
420 end
421
422 end
423
424 %% =====
425 % 13. ENSURE FINAL VALUES ARE RECORDED
426 % =====
427
428 omega(end) = x(2);
429 RPM(end) = omega(end)*60/(2*pi);
430
431 %% =====
432 % 14. SAFETY CHECK
433 % =====
434
435 % Find the first instant at which the PLC permits cylinder actuation.
436
437 idx_release = find(Cylinder_PLC > 0,1,'first');
438
439 if ~isempty(idx_release)
440
441     release_time = t(idx_release);
442     release_RPM = RPM(idx_release);
443
444 else
445
446     release_time = NaN;
447     release_RPM = NaN;
448
449 end
450
451 %% =====
452 % 15. UNSAFE RELEASE CHECK
453 % =====
454
455 idx_nonPLC = find(Cylinder_nonPLC > 0,1,'first');
456
457 unsafe_release_time = t(idx_nonPLC);
458 unsafe_release_RPM = RPM(idx_nonPLC);
459
460 %% =====

```

```

461 % 16. DISPLAY RESULTS
462 % =====
463
464 fprintf('\n');
465 fprintf('=====\\n');
466 fprintf(' CNC SPINDLE TOOL-CHANGE INTERLOCK SIMULATION\\n');
467 fprintf('=====\\n');
468
469 fprintf('\\nInitial spindle speed: %.1f RPM\\n', RPM_nominal);
470
471 fprintf('\\nWITHOUT PLC INTERLOCK:\\n');
472 fprintf('Cylinder release time: %.4f s\\n', unsafe_release_time);
473 fprintf('Spindle speed at release: %.2f RPM\\n', unsafe_release_RPM);
474
475 fprintf('\\nWITH PLC INTERLOCK:\\n');
476 fprintf('Cylinder release time: %.4f s\\n', release_time);
477 fprintf('Spindle speed at release: %.2f RPM\\n', release_RPM);
478
479 fprintf('\\nStopping threshold: %.2f RPM\\n', RPM_stop_threshold);
480
481 % Extra explicit safety assertion for the report
482 if ~isnan(release_RPM)
483     if abs(release_RPM) <= RPM_stop_threshold
484         fprintf('Safety check: PASS - PLC release speed is within threshold.\\n');
485     else
486         fprintf('Safety check: FAIL - PLC release speed is above threshold.\\n');
487     end
488 end
489
490 %% =====
491 % 17. FIGURE 1 - SPINDLE SPEED AND TOOL-CHANGE REQUEST
492 % =====
493
494 figure('Name','Spindle Speed During Tool Change');
495
496 yyaxis left
497
498 plot(t,RPM,'LineWidth',1.5);
499 hold on;
500
501 % Filtered reference (post-LPF) is what the PID actually tracks.
502 plot(t,omega_ref_used * 60/(2*pi),'--','LineWidth',1.2);
503
504 yline(RPM_stop_threshold,':','Stopping Threshold');
505
506 ylabel('Spindle Speed (RPM)');
507

```

```

508     yyaxis right
509
510     plot(t,Tool_Change_Request,'LineWidth',1.2);
511
512     ylabel('Tool Change Request');
513
514     xlabel('Time (s)');
515     title('CNC Spindle Speed During Tool-Change Sequence');
516
517     grid on;
518
519     legend('Actual RPM',...
520           'PLC Commanded RPM (filtered)',...
521           'Stopping Threshold',...
522           'Tool Change Request',...
523           'Location','best');
524
525     %% =====
526     % 18. FIGURE 2 - NON-PLC VS PLC CYLINDER ACTUATION
527     % =====
528
529     figure('Name','PLC Interlock Comparison');
530
531     plot(t,RPM/1000,'LineWidth',1.5);
532     hold on;
533
534     plot(t,Cylinder_nonPLC*5,'--','LineWidth',1.5);
535
536     plot(t,Cylinder_PLC*5,'-.','LineWidth',1.5);
537
538     yline(RPM_stop_threshold/1000,':');
539
540     xlabel('Time (s)');
541     ylabel('Speed / Actuator State');
542
543     title('Comparison of Tool-Release Operation');
544
545     legend('Spindle Speed (1000 RPM)',...
546           'Cylinder - No PLC',...
547           'Cylinder - PLC Interlock',...
548           'Stopping Threshold',...
549           'Location','best');
550
551     grid on;
552
553     %% =====
554     % 19. FIGURE 3 - SAFETY TIMELINE (5 PANELS)

```

```

555 %
556 % Extra panel added: the PLC Stop Command is now shown explicitly so the
557 % reader can see the supervisory command separate from the cylinder firing.
558 % =====
559
560 figure('Name','Tool Change Safety Timeline');
561
562 subplot(5,1,1);
563 plot(t,RPM,'LineWidth',1.5);
564 hold on;
565 yline(RPM_stop_threshold,':');
566 ylabel('RPM');
567 title('Tool-Change Safety Sequence');
568 grid on;
569
570 subplot(5,1,2);
571 stairs(t,Tool_Change_Request,'LineWidth',1.5);
572 ylim([-0.1 1.1]);
573 ylabel('Request');
574 grid on;
575
576 subplot(5,1,3);
577 stairs(t,PLC_Stop_Command,'LineWidth',1.5);
578 ylim([-0.1 1.1]);
579 ylabel('PLC Stop');
580 grid on;
581
582 subplot(5,1,4);
583 stairs(t,Cylinder_nonPLC,'LineWidth',1.5);
584 ylim([-0.1 1.1]);
585 ylabel('No PLC');
586 grid on;
587
588 subplot(5,1,5);
589 stairs(t,Cylinder_PLC,'LineWidth',1.5);
590 ylim([-0.1 1.1]);
591 ylabel('PLC');
592 xlabel('Time (s)');
593 grid on;
594
595 %% =====
596 % 20. FIGURE 4 - PLC STATE SEQUENCE
597 %
598 % The state sequence is now taken directly from PLC_State_signal, which
599 % is logged during the simulation. This guarantees the plot reflects the
600 % actual state machine execution (no post-hoc reconstruction that could
601 % disagree with the logic).

```

```
602 % =====
603
604 figure('Name','PLC Supervisory State Sequence');
605
606 stairs(t,PLC_State_signal,'LineWidth',1.8);
607
608 yticks([0 1 2 3]);
609
610 yticklabels({'NORMAL',...
611             'STOPPING',...
612             'RELEASE',...
613             'COMPLETE'});
614
615 xlabel('Time (s)');
616 ylabel('PLC State');
617
618 title('Supervisory PLC State Sequence');
619
620 grid on;
621
622 %% =====
623 % 21. SAVE FIGURES FOR REPORT
624 % =====
625
626 saveas.figure(1),'Pictures/spindle_tool_change_speed.png');
627 saveas.figure(2),'Pictures/plc_interlock_comparison.png');
628 saveas.figure(3),'Pictures/tool_change_safety_timeline.png');
629 saveas.figure(4),'Pictures/plc_state_sequence.png');
630
631 %% =====
632 % END OF SCRIPT
633 % =====
```

MATLAB/Simulink Block Diagrams

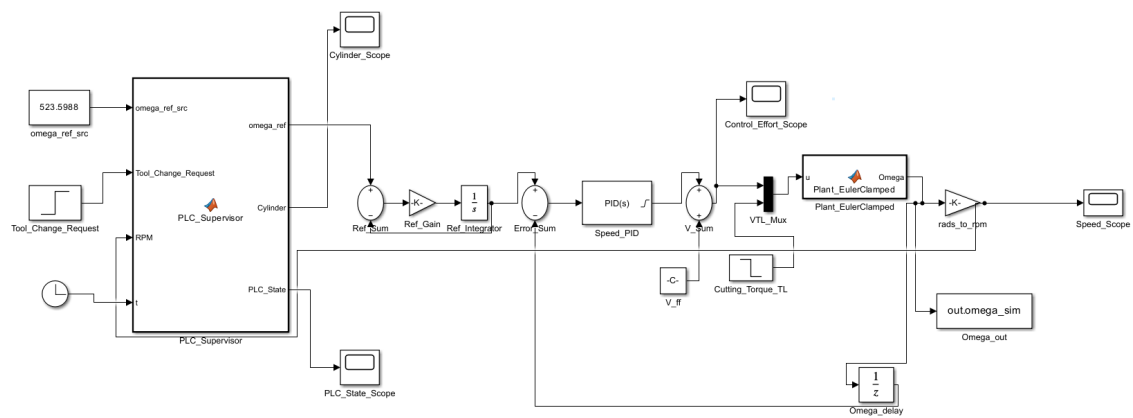


Figure C.1: MATLAB/Simulink block diagram.